Audio Equipment and Systems

Bachelor's Degree in Audiovisual Systems Engineering

Sound Capturing and Microphones

Engineering School
Department of Information
and Communication Technologies

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Sound Capturing and Microphones

- Transducer classification of microphones
  - Electrodynamic microphones
    - Moving-coil transduction
    - Ribbon transduction
  - Piezoelectric microphones
  - Electrostatic microphones
  - Electret technology
  - MEMS microphones
- Acoustic classification of microphones
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- Impedance
- Electroacoustic analogies
- Phantom power
- DI's
- Balanced vs non-balanced
- Wireless transmission
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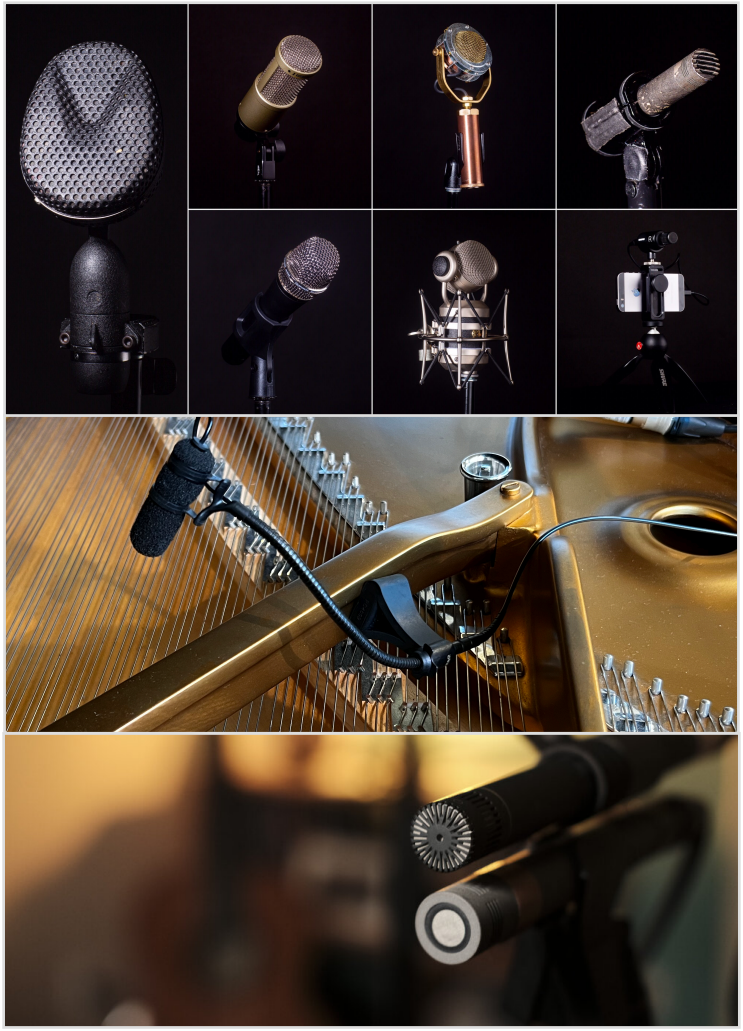
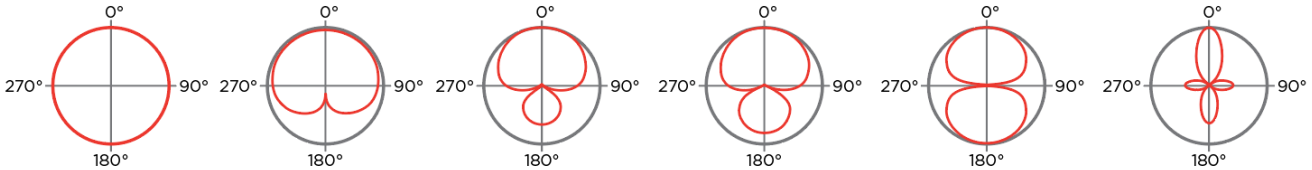
- Recording case studies



Classification of microphones

- Transduction principle
  - Electrodynamic microphones (passive)
    - Moving coil
    - Ribbon
  - Electrostatic microphones (active)
    - Condenser
    - Electret
  - Piezoelectric microphones (active or passive)

- Acoustic classification (electrodynamic or electrostatic /active or passive)



  - Pressure microphones => Omnidirectional response
  - Gradient pressure microphones => Directional response
    - Cardioid response
    - Figure of eight response
    - Super/Hyper cardioids
- Special applications

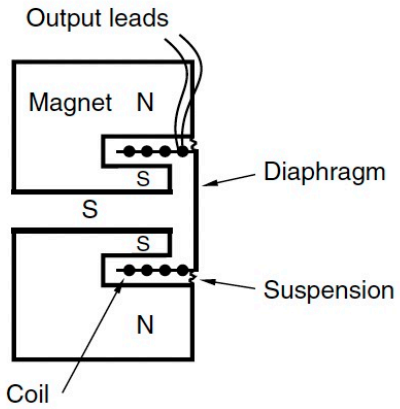
Passive => Don't need power supply
Active => Need power supply (phantom/batteries)

Electrodynamic Microphones

FACT FILE 3.2 DYNAMIC MICROPHONE – PRINCIPLES

The moving-coil microphone functions like a moving-coil speaker in reverse. As shown in the diagram, it consists of a rigid diaphragm, typically 20–30 mm in diameter, which is suspended in front of a magnet. A cylindrical former is attached to the diaphragm on to which is wound a coil of very fine-gauge wire. This sits in the gap of a strong permanent magnet. When the diaphragm is made to vibrate by sound waves the coil in turn moves to and fro in the magnet's gap, and an alternating current flows in the coil, producing the electrical output (see Fact File 3.1). Some models have sufficient windings on the coil to produce a high enough output to be fed directly to the output terminals, whereas other models use fewer windings, the lower output then being fed to a step-up transformer in the microphone casing and then to the output. The resonant frequency of dynamic microphone diaphragms tends to be in the middle frequency region.

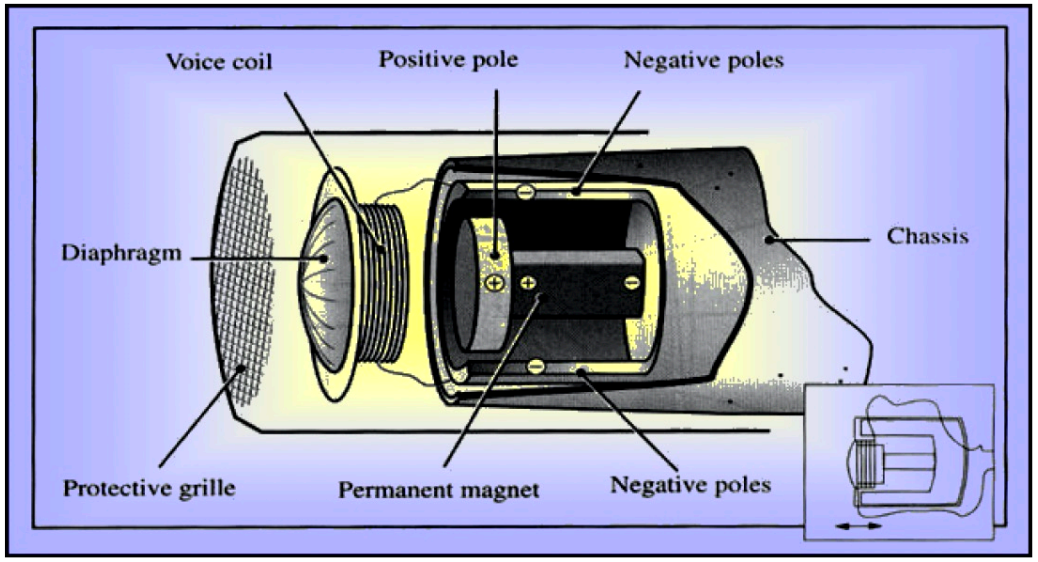
The standard output impedance of professional microphones is 200 ohms. This value was chosen because it is high enough to allow useful step-up ratios to be employed in the output transformers, but low enough to allow a microphone to drive long lines of 100 meters or so. It is possible, though, to encounter dynamic microphones with output impedances between 50 and 600 ohms. Some moving-coil models have a transformer that can be wired to give a high-level, high-impedance output suitable for feeding into the lower-sensitivity inputs found on guitar amplifiers and some PA amplifiers. High-impedance outputs can, however, only be used to drive cables of a few meters in length, otherwise severe high-frequency loss results. (This is dealt with fully in Chapter 12.)



RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012, p 49.

How microphones work: https://www.youtube.com/watch?v=Bt7vXL1_Y1Y&t=4s

Moving coil transduction

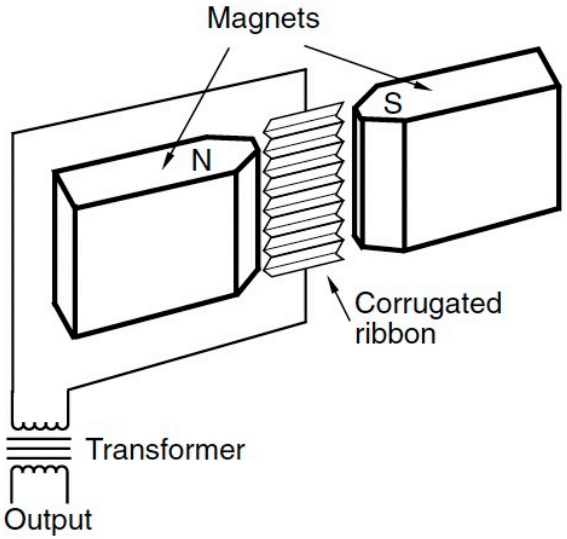


pros	cons
reliabilty, strong	transients slow response
supports over-pressure (“hard” mic)	low sensitivity
coloured/selective frequency response	Non-flat (tailored-response) BW
no *phantom power*	
low output impedance	

Ribbon Microphones

FACT FILE 3.3 RIBBON MICROPHONE – PRINCIPLES

The ribbon microphone consists of a long thin strip of conductive metal foil, pleated to give it rigidity and 'spring', lightly tensioned between two end clamps, as shown in the diagram. The opposing magnetic poles create a magnetic field across the ribbon such that when it is excited by sound waves a current is induced into it (see Fact File 3.1). The electrical output of the ribbon is very small, and a transformer is built into the microphone which steps up the output. The step-up ratio of a particular ribbon design is chosen so that the resulting output impedance is the standard 200 ohms, this also giving an electrical output level comparable with that of moving-coil microphones. The resonant frequency of ribbon microphones is normally at the bottom of the audio spectrum.



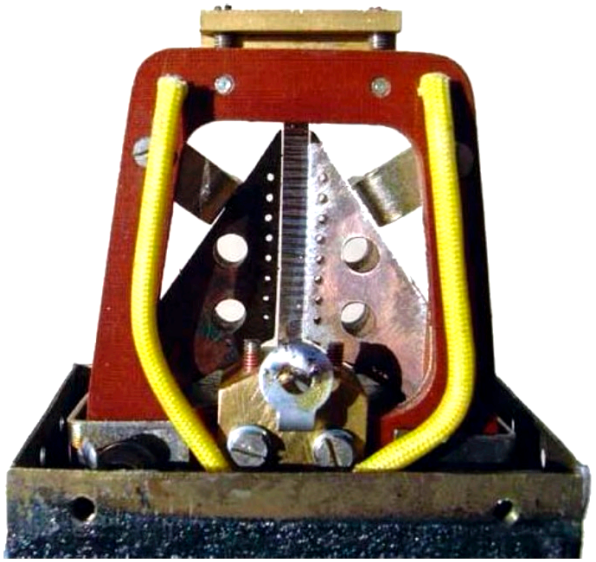
RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012, p 49.

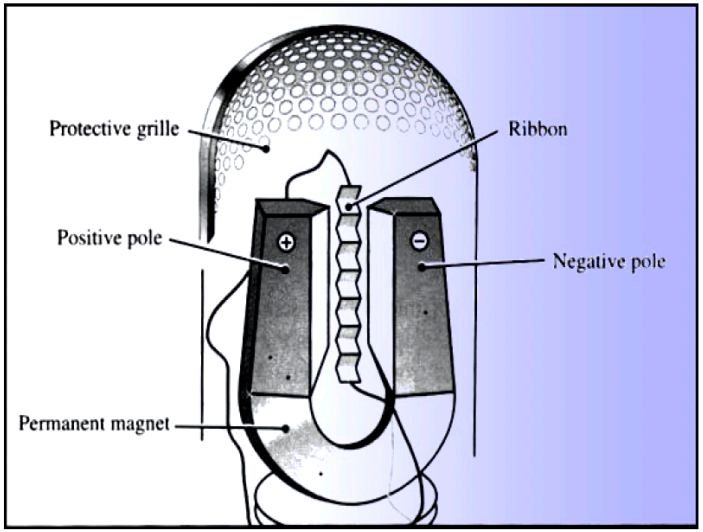
Ribbon transduction

How ribbon microphones are made:

https://aearibbonmics.com/our-story/

pros	cons
sensitivity	fragile
extended BW	price
coloured BW response	coloured BW response
transient fast response	_
low output impedance	_
no *phantom power*	be careful with *phanthom!*
personality / colour	_

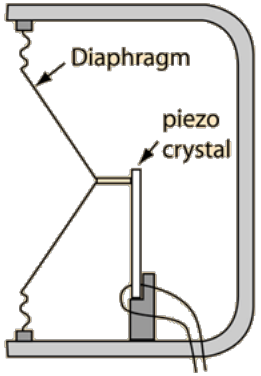


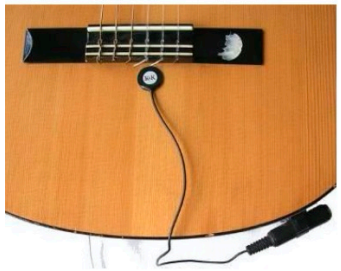


Piezoelectric transduction

- Piezoelectricity (Pierre Curie, 1880): certain crystalline or ceramic materials such as Rochelle salts, $BaTiO_3$ (Barium titanate), or ADP (ammonium di-hydrogen phosphate), generate a voltage when flexed or deformed, and vice-versa.

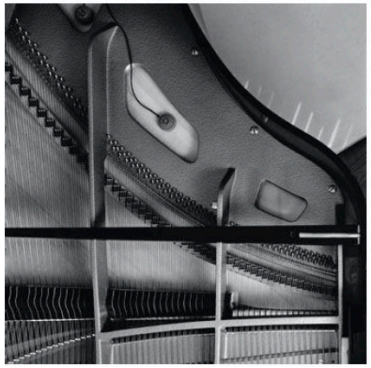
  Examples: Acoustic guitar pick-up, contact microphone





pros	cons
no feedback	
poor sensitivity	poor sensitivity
simple & strong construction	reduced freq. BW
sort of "invisible"	high output impedance

Headway The Band



AKG DB-1

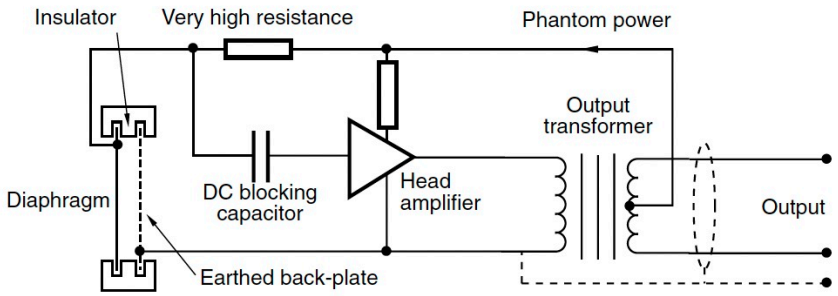
Electrostatic Microphones

FACT FILE 3.4 CAPACITOR MICROPHONE – PRINCIPLES

The capacitor (or condenser) microphone operates on the principle that if one plate of a capacitor is free to move with respect to the other, then the capacitance (the ability to hold electrical charge) will vary. As shown in the diagram, the capacitor consists of a flexible diaphragm and a rigid back plate, separated by an insulator, the diaphragm being free to move in sympathy with sound waves incident upon it. The 48 volts DC phantom power (see 'Microphone powering options', below) charges the capacitor via a very high resistance. A DC blocking capacitor simply prevents the phantom power from entering the head amplifier, allowing only audio signals to pass.

When sound waves move the diaphragm the capacitance varies, and thus the voltage across the capacitor varies proportionally, since the high resistance only allows very slow leakage of charge from the diaphragm (much slower than the rate of change caused by audio frequencies). This voltage modulation is fed to the head amplifier (via the blocking capacitor) which converts the very high impedance output of the capacitor capsule to a much lower impedance. The output transformer balances this signal (see 'Balanced lines', Chapter 12) and conveys it to the microphone's output terminals. The resonant frequency of a capacitor mic diaphragm is normally at the upper end of the audio spectrum.

The head amplifier consists of a field-effect transistor (FET) which has an almost infinitely high input impedance. Other electronic components are also usually present which perform tasks such as voltage regulation and output stage duties. Earlier capacitor microphones had valves built into the housing, and were somewhat more bulky affairs than their modern counterparts. Additionally, extra wiring had to be incorporated in the mic leads to supply the valves with HT (high-tension) and valve-heater voltages. They were thus not particularly convenient to use, but such is the quality of sound available from capacitor mics that they quickly established themselves. Today, the capacitor microphone is the standard top-quality type; other types being used for relatively specialized applications. The electrical current requirement of capacitor microphones varies from model to model, but generally lies between 0.5 mA and 8 mA, drawn from the phantom power supply.



RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012, p 50.

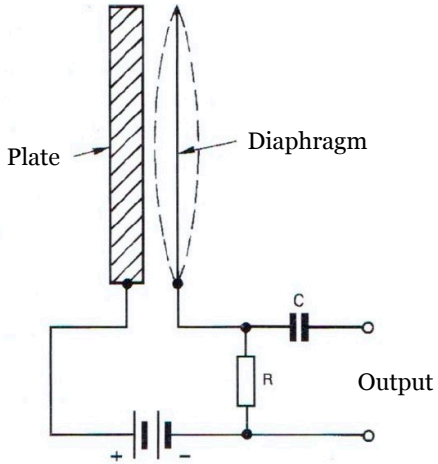
Electrostatic transduction

How are made? https://www.youtube.com/watch?v=A4lMkKnz6Ts

$$Sensitivity \approx \frac{E_{0*}a^2}{h*8*T_0} \; V/Pa$$

- capacitor microphone (Western Electric, 1917)

AKG C414 (1962-2009)



Polarization voltage

pros	cons
high sensitivity	fragile
Broad BW response	price
flat BW response	*Phantom* needed
fast transient response	
low output impedance	

characteristics	typical physical values
polarizing voltage E_0	48-52V (up to 300V)
backplate-diaphragm distance h	25µm
diaphragm radius a	10-50mm
diaphragm mass density σ_m	0.05kg/m^2
diaphragm capacity	15-50pF
parallel resistance	$10^9\Omega$
effective diaphragm tension T_o	2000N/m
sensitivity range	5-50mV/Pa

Neumann CMV-3 (1928)

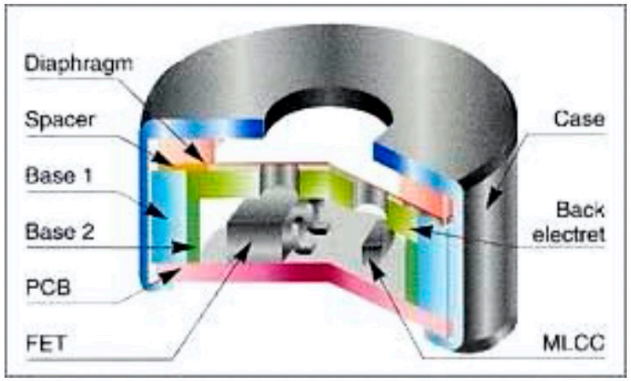
vintage designs renewed…

Blue Microphones The Bottle (≈2000)

Electret technology

- electret capacitor microphone (~1960s)
  - prepolarized *Backplate* with permanent electrostatic charge from ferroelectric materials ($BaTiO_3$)
  - a constant electric field supplies permanent electrostatic charge Q

pros	cons
sensitivity	fragile
"invisible"	reduced BW
flat frequency. response	noise level
no *phantom*	
low output impedance	



MEMS microphones

Micro electro-mechanical systems

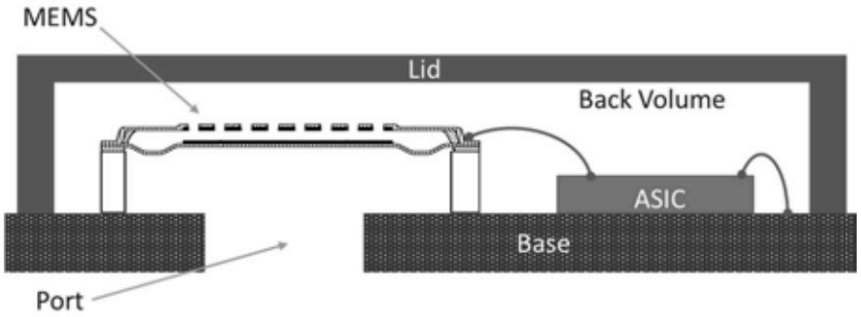


Figure 1: The microphone without metal lid (top) and a schematic cross-sectional view of the microphone (bottom).

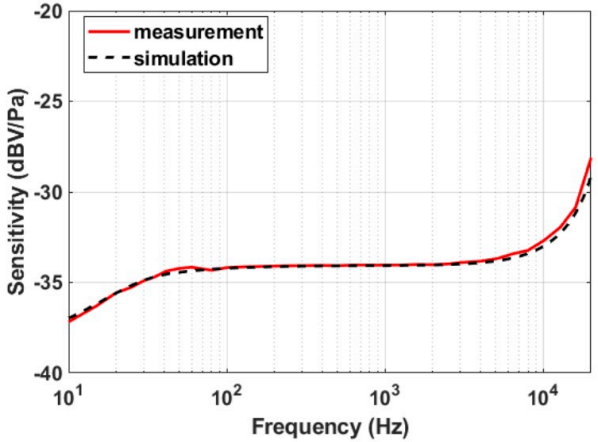


Figure 7: Measured and simulated frequency response.

MEMS compared to Electret Condenser

- Are smaller
- Can provide ADC incorporated in ASIC (Application specific integrated circuit)
- Have lower impedance (better in electrically noisy environments)
- Have a narrower operating voltage range
- Are highly tolerant to external mechanical vibration and extreme temperature variations

NADERYAN, Vahid, et al. Mems microphone with 73dba snr in a 4mm x 3mm x 1.2 mm package. En *2021 21st International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers)*. IEEE, 2021. p. 242-245.

Transducer classification. Comparative

Mic Type	Advantages	Drawbacks	Frequency Response	Transient Response
Electrodynamic Moving Coil	Durable, rugged, well-suited for live and high SPL sources (e.g., drums, guitar amps). No external power needed.	Less sensitive, not ideal for studio detail recordings.	Generally limited high-frequency extension; strong midrange presence.	Moderate; good for loud sources but less precise on rapid transients.
Electrodynamic Ribbon	Natural, warm tone close to human hearing, excellent for vocals and instruments.	Fragile, delicate, can be damaged by phantom power, and often more expensive than moving coil mics.	Smooth, rolled-off highs giving a warm, natural sound; less boosted highs.	Excellent; very fast and accurate transient response due to light moving element.
Electrostatic Condenser	High sensitivity, excellent high-frequency response, ideal for voice and studio recording. Also used in live environments for sources like piano or percussion. Wide polar pattern options.	Fragile, needs phantom power, more expensive, less durable in rough handling.	Wide and extended frequency range, typically up to 20 kHz+ with detailed highs.	Very fast; highly capable of capturing quick transients with clarity.
Electrostatic Electret	Compact, low cost, good for handheld, portable applications, readily available in consumer devices. Work with small batteries.	Lower audio fidelity compared to true condensers, can be less durable over time.	Generally wide but less extended and less accurate than true condensers.	Moderate; decent transient response, but less precise for rapid changes in sound.
Piezoelectric	Suitable for contact and surface-mount applications, small size, good for sensing vibrations or specific acoustic applications.	Generally poor frequency response for high-quality sound, limited to specialized uses.	Narrow frequency range, often focused on low to mid frequencies.	Generally slow; suitable for static or low-frequency vibrations, not quick transients.
MEM	Tiny, embedded in consumer electronics, reliable small sensors, and microphones integrated into devices like smartphones.	Limited audio quality, mainly designed for speech/voice recognition, not suitable for high-fidelity audio.	Limited frequency range compared to professional mics, typically 100 Hz to 8 kHz approx.	Slow to moderate; not optimized for transient accuracy but sufficient for speech.

Transducer classification. Applications

Microphone Type	Typical Applications	Power required
Moving Coil	Live vocals, live music instruments (guitar amps, drums), broadcasting, podcasting, stage performances, public speaking. Suitable for high SPL and rugged environments.	No (passive)
Ribbon	Studio vocals, guitar amplifiers, orchestral and chamber music recordings, brass and string instruments, ambient and field recording where a warm natural tone is preferred.	No (passive)
Condenser	Studio recordings of vocals and instruments, drum overheads and acoustic guitars, piano recordings, classical and jazz ensembles, voiceover work, broadcasting, and high-fidelity capture.	Yes (phantom power)
Electret	Portable and wearable devices (lavalier mics), smartphones, tablets, conferencing systems, consumer audio devices, voice assistants.	Yes (batteries/phantom power)
Piezoelectric	Contact pickups for acoustic instruments (guitar, violin), vibration sensors in musical instruments, industrial and medical acoustic sensing applications, specialty audio pickups.	No (passive)
MEM	Embedded in consumer electronics such as smartphones, laptops, hearing aids, wireless earbuds, smart speakers, voice-activated devices, and speech recognition systems.	Yes (integrated preamp/ ADC)

Technical rider. Track-list

Exercise: In live concerts it is a common practice to suggest your microphone preferences and technical needs in a document called rider. Search on the internet one microphone for each instrument in the following track-list of a jazz ensemble.

Channel	Instrument	Example	Your list (Brand and model)
1	Kick drum	AKG D112	
2	Snare drum	Shure SM 57	
3-4	Drum overhead L-R (cymbals, overall kit ambience)	2x Neumann KM 184	
5	Double bass (acoustic)	DPA 4099	
5	Electric guitar (amplifier)	Shure SM57	
7-8	Piano (L-R)	2x DPA 4099	
9	Trumpet	Shure SM 58	
10	Trombone	Sennheiser MD421	
11	Saxophone	AKG 414	
12	Vocals (lead singer)	Neumann KMS 105	
13-14	Backing vocals	Shure SM58	

Some commercial examples

Sennheiser MD 421-II
(Dynamic - Coil)

Shure KSM313/NE
(Dynamic - Ribbon)

Neumann KM 184 - series
(Electrostatic - Condenser)

Acoustic classification. First order patterns

- Random Efficiency (RE/REE)
  Power on-axis vs all directions
- Directivity index (DI [dB])
  Power on-axis vs all directions
- Distance factor (DF)
  distance to preserve direct-to-reverberant pickup

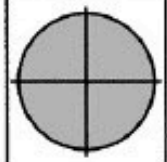
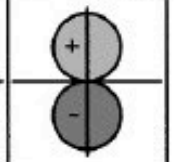
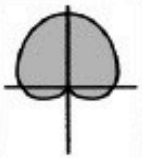
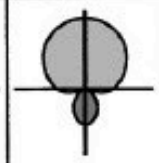
CHARACTERISTIC	PRESSURE COMPONENT	GRADIENT COMPONENT	SUBCARDIOID	CARDIOID	SUPERCARDIOID	HYPERCARDIOID
POLAR RESPONSE PATTERN						
POLAR EQUATION	1	cos θ	.7 + .3 cos θ	.5 + .5 cos θ	.37 + .63 cos θ	.25 + .75 cos θ
PICKUP ARC 3 dB DOWN	360°	90°	180°	131°	115°	105°
PICKUP ARC 6 dB DOWN	360°	120°	264°	180°	156°	141°
RELATIVE OUTPUT AT 90° (dB)	0	−∞	-3	-6	-8.6	-12
RELATIVE OUTPUT AT 180° (dB)	0	0	-8	−∞	-11.7	-6
ANGLE AT WHICH OUTPUT = ZERO	–	90°	–	180°	126°	110°
RANDOM EFFICIENCY (RE)	1	.333	.55	.333	.268(1)	.25(2)
DIRECTIVITY INDEX (DI)	0 dB	4.8 dB	2.5 dB	4.8 dB	5.7 dB	6 dB
DISTANCE FACTOR (DSF)	1	1.7	1.3	1.7	1.9	2

Figure 5.4
Characteristics of the family of first-order cardioid microphones.

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, pp 85, 86.

Distance and directivity factors

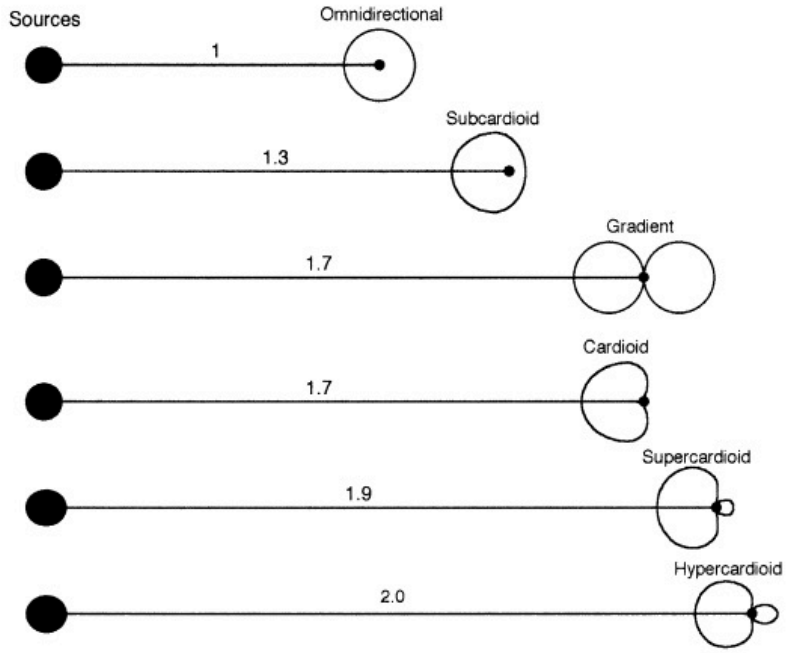


Figure 5.5
Illustration of distance factor (DF) for the first-order cardioid family.

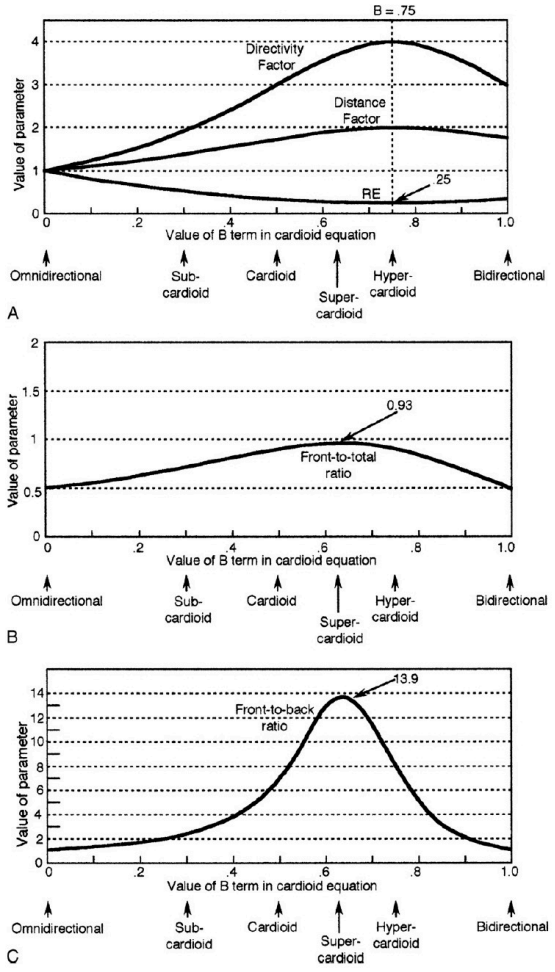


Figure 5.6
Graphical illustrations of directivity factor, distance factor, random efficiency, front-to-total ratio, and front-to-back ratio for the first order family.

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, pp 87, 88.

Cardioid Condenser transducer

https://www.animations.physics.unsw.edu.au/jw/compliance-inertance-impedance.htm

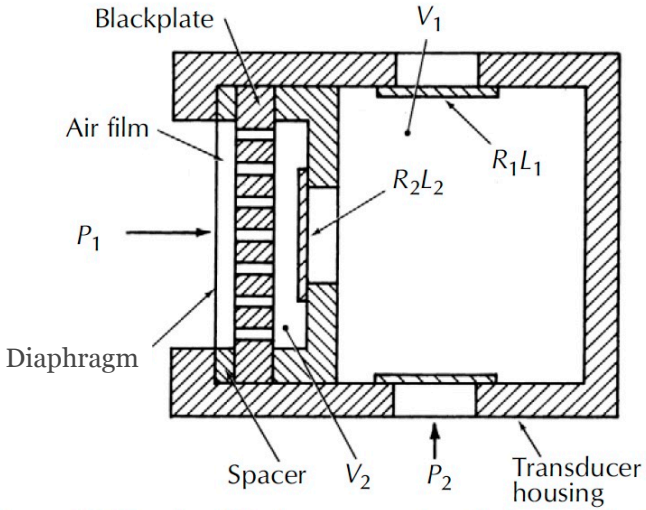


Figure 16-22. Simplified cross-sectional view of the Shure SM81 condenser transducer. Courtesy Shure Incorporated.

P1: front pressure
P2: rear entry
Z0: diaphragm acoustical impedance
R1-L1 & R2-L2: Resistances and inertances (screen mounted in the side windows)
V1 & V2: cavity volumes
C1 & C2: cavity compliances

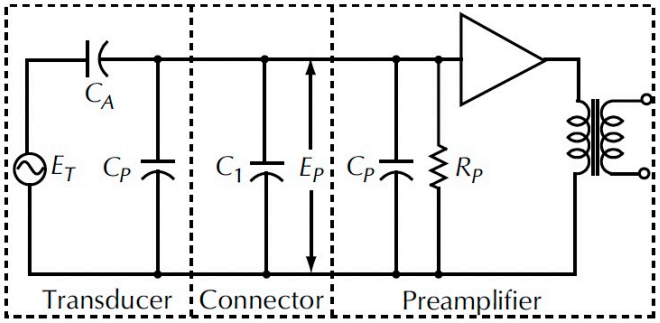


Figure 16-23. Electrical equivalent circuit of the Shure SM81 condenser transducer and preamplifier. Courtesy Shure Incorporated.

SHURE SM 81 SPECS. Courtesy Shure Incorporated.

Transducer Type: Condenser
Polar Pattern: Cardioid
Frequency Range: 20Hz-20KHz
Sensitivity: -45.00 dBV/Pa
Sensitivity: 5.60 mv/pa
Equivalent Selfnoise: 16 dB(A)
Maximum Sound Pressure
800 ohm load: 136 dB (attenuator at 0)
150 ohm load: 128 dB (attenuator at 0)
Impedance: 85 Ω

BALLOU, Glen. *Handbook for sound engineers.* Taylor & Francis, 2013, p 500.

Cardioid Ribbon transducer

Pure Ribbon transducers are bidirectional but it is possible to provide them with cardioid characteristics:

- A portion of the ribbon is exposed to the sound field on both sides
- An acoustic labyrinth (delay) at the rear of the other ribbon portion can be tuned to adjust the response

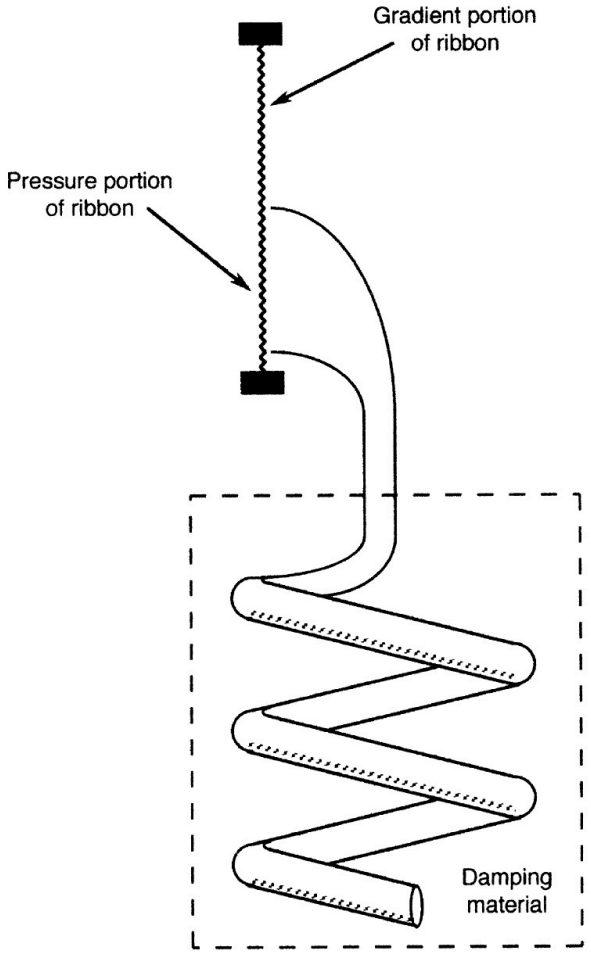
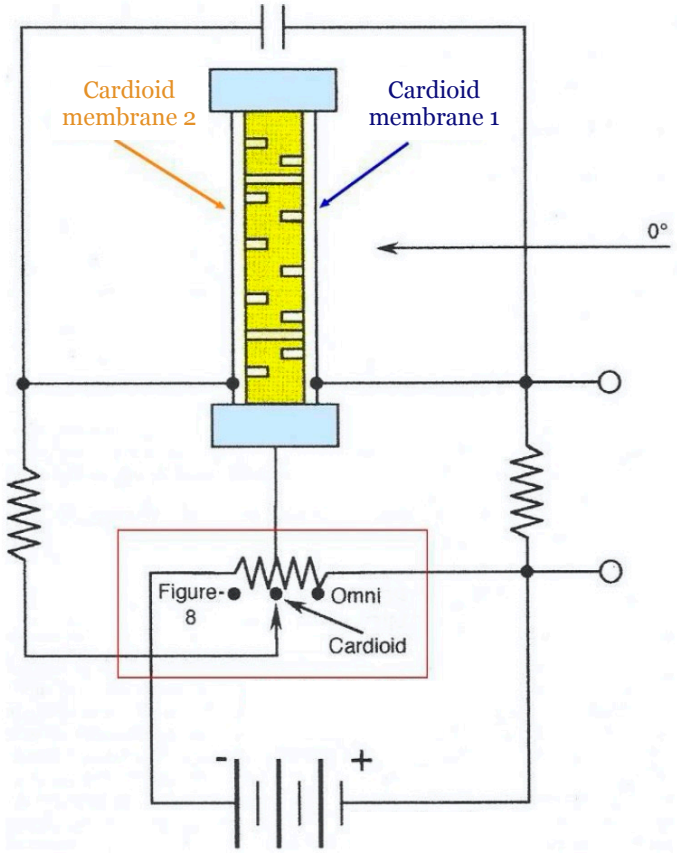


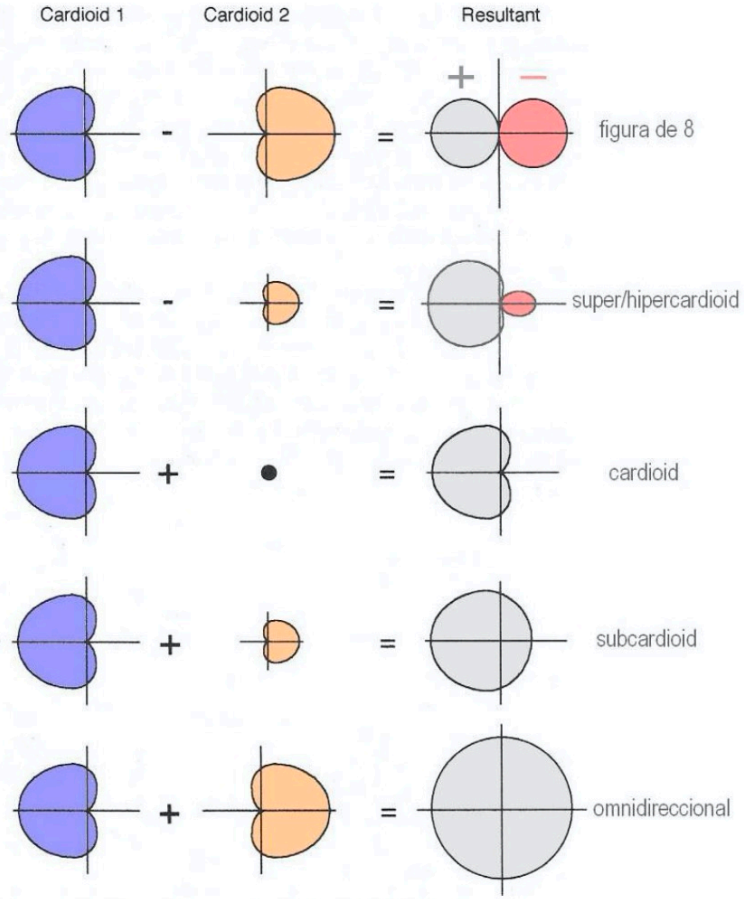
Figure 5.9
Section view of the mechanism of a ribbon cardioid microphone.

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, pp 85, 86.

Dual diaphragm microphones

- dual diaphragm pressure gradient circuit (Braunmühl-Weber, 1935)
  - as in microphones like Audiotechnica 4050, AKG 414, Neumann U47/U48, U87, etc.



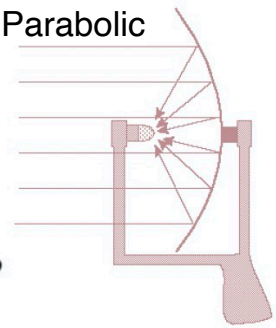


Special applications

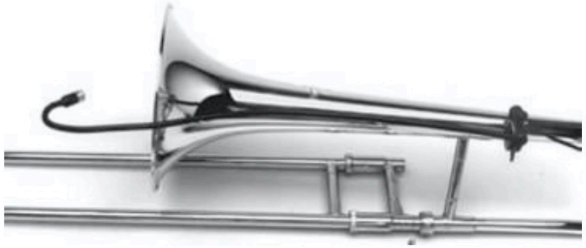
- clip-on, lavalier
- parabolic
- shotgun
- specialized

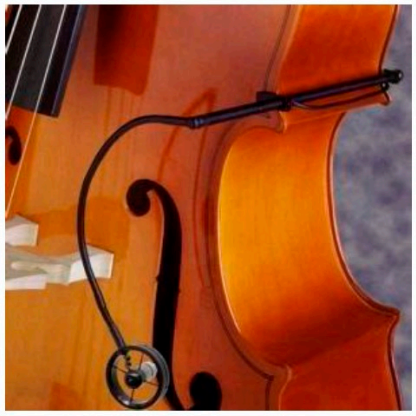
AT8471

Parabolic

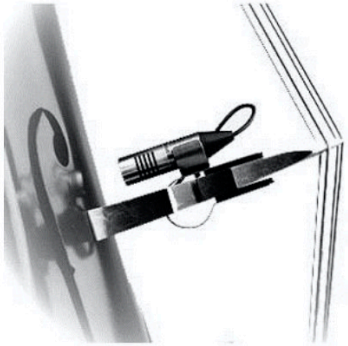


Sub-kick



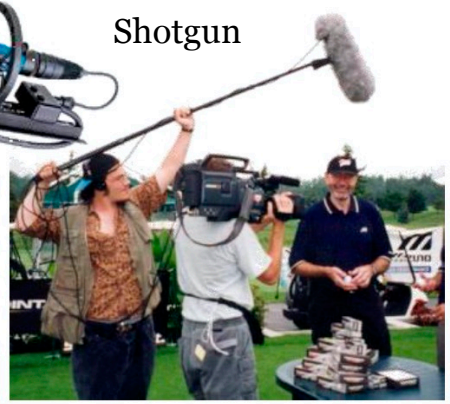


Miniatures



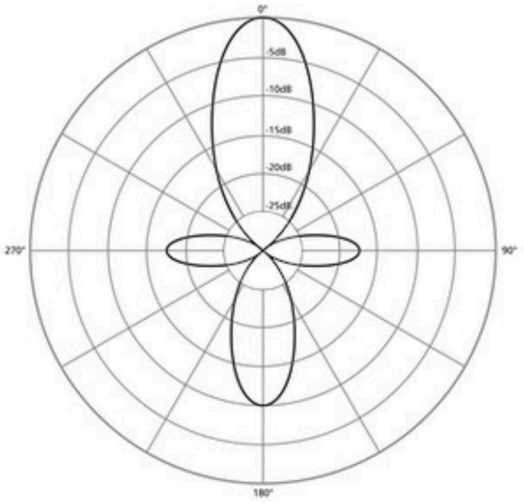
Clip

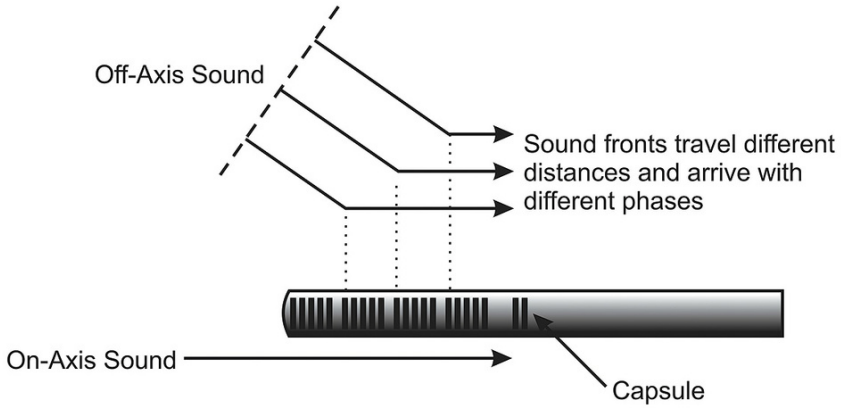
Shotgun

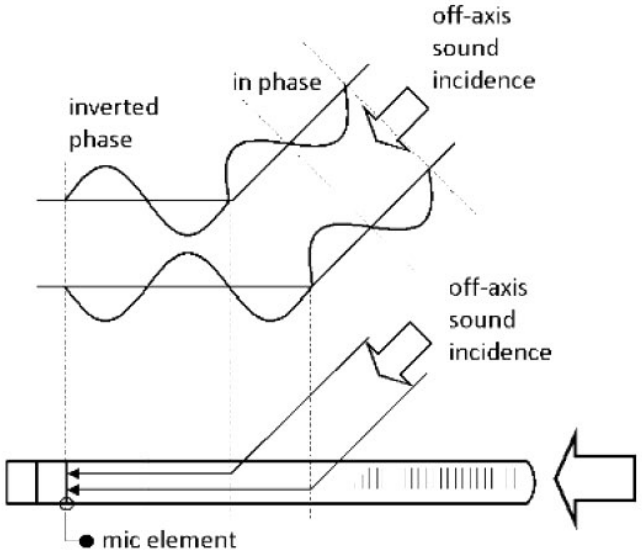


Interference tube. Shotgun Mics

Application: Broadcast, TV and film sound recording







On-Axis sound front follows a straight path to the membrane

Off-Axis sound front experiments destructive interferences

https://www.dpamicrophones.com/mic-university/technology/the-interference-tube-and-its-use-in-microphones/

Multichannel capturing

Neumann USM 69 I

Schoeps STC 4 with 2 CCM4
(Cardioid)

Schoeps with 4 CCM41
and 4 CCM41 V (3D)

Schoeps LM with 4 CCM41
(Supercardioid)

Stereo capturing techniques

Technique	Microphone Type	Spacing/Angle	Advantages	Drawbacks	Typical Use
X-Y (Coincident)	2 Cardioid	Coincident/capsules close. 90°–135°	Excellent phase coherence, mono compatibility	Narrow stereo image	Small ensembles, solo instrument
A-B (Spaced Pair)	2 Omni or Cardioid	Parallel, spaced (20–35 cm or more)	Wide stereo image, strong sense of space	May cause phase issues, poor mono	Orchestras, choirs, ambiance
M-S (Mid-Side)	1 Cardioid (Mid), 1 Fig-8 (Side)	Coincident, capsules aligned; Mid faces source, Side 90°	Adjustable width, excellent mono compatibility	Requires decoding, sensitive setup	Broadcast, classical, field recording
ORTF (Near Coincident)	2 Cardioid	17 cm spacing, 110° angle	Combines timing/ intensity cues; wide image	Center source slightly off-axis	Ensemble, piano, natural room sound
DIN (Near Coincident)	2 Cardioid	20 cm spacing, 90° angle	Blends delay/intensity; good room ambience	Low-frequency loss at distance	Small ensemble, choir
Blumlein Pair (Coincident)	2 Fig-8	Coincident/capsules close. 90°	Realistic room/space; great for acoustic spaces	Sensitive to room; less direct sound	Room/ambiance, orchestras
Jecklin Disk (Spaced Pair)	2 Omni	16.5 cm apart, separated by disk	Natural, spatial sound; good localization	Bulky setup; headphone-centric sound	Orchestras, choirs, binaural

MS (Mid-Side) signals

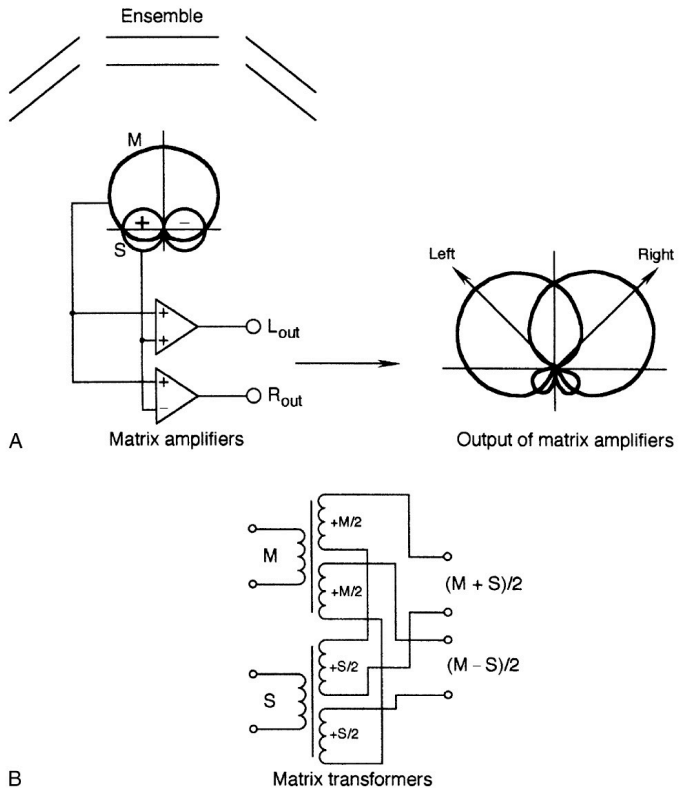


Figure 11.7
M-S system: a front-facing cardioid and side-facing figure-8, when mixed as shown, are equivalent to a pair of splayed cardioid microphones (A); alternate method of achieving sum and difference outputs using transformers (B).

EARGLE, John.
The Microphone Book.
From mono to stereo to surround -a guide to microphone design and application.
CRC Press, 2012, p 223..

FACT FILE 3.6 SUM AND DIFFERENCE PROCESSING

MS signals may be converted to conventional stereo very easily, either using three channels on a mixer, or using an electrical matrix. M is the mono sum of two conventional stereo channels, and S is the difference between them. Thus:

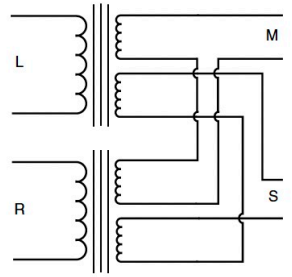
$$M = (L + R) \div 2$$
$$S = (L - R) \div 2$$

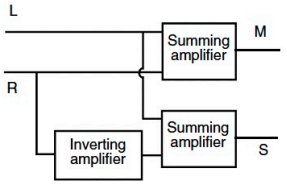
and

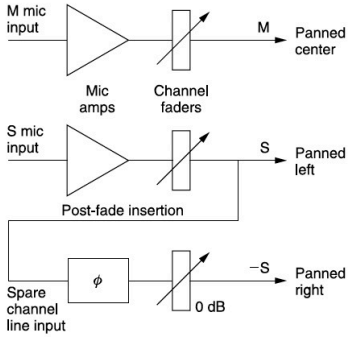
$$L = (M + S) \div 2$$
$$R = (M - S) \div 2$$

A pair of transformers may be used wired as shown in the diagram to obtain either MS from LR, or vice versa. Alternatively, a pair of summing amplifiers may be used, with the M and S (or L and R) inputs to one being wired in phase (so that they add) and to the other out of phase (so that they subtract).

The mixer configuration shown in the diagram may also be used. Here the M signal is panned centrally (feeding L and R outputs), whilst the S signal is panned left (M + S = L). A post-fader insertion feed is taken from the S channel to a third channel which is phase reversed to give −S. The gain of this channel is set at 0dB and is panned right (M − S = R). If the S fader is varied in level, the width of the stereo image and the amount of rear pickup can be varied.





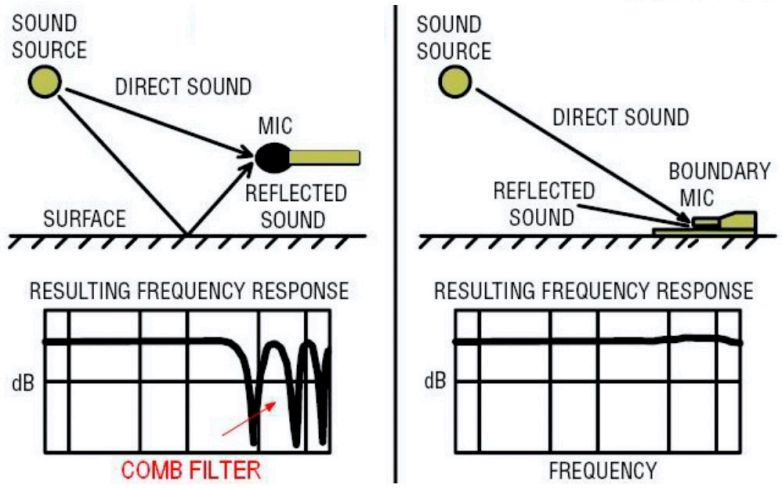


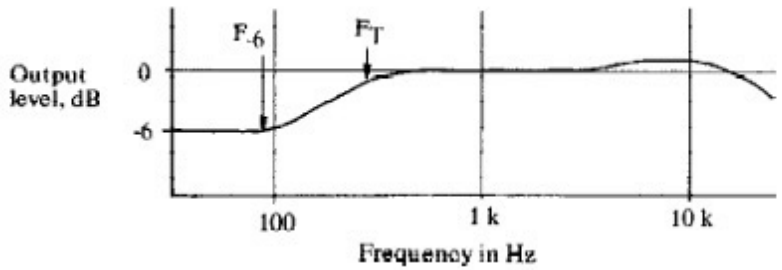
RUMSEY, Francis; MCCORMICK, Tim.
Sound and recording: an introduction.
CRC Press, 2012, p 63.

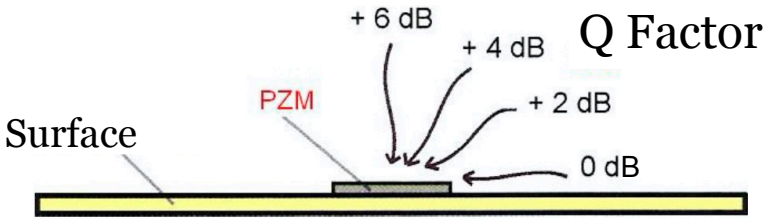
Special applications. Boundary Mics

Application: Theater on-stage, kick drum, etc..

- PZM (Pressure Zone Mode)







Q Factor

$F_T = 750/A$

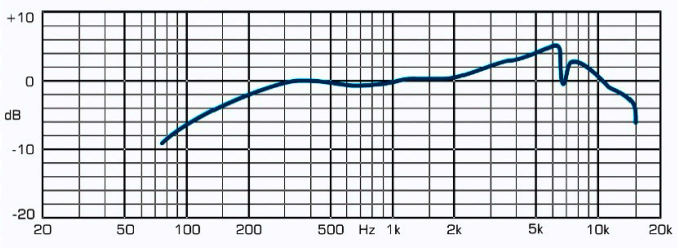
$F_{-6} = 188/A$

A: plate area [feet squared]

https://www.soundonsound.com/sound-advice/q-pzm-and-boundary-mics-whats-difference

Technical specifications

- **sensitivity** → mV for 1Pa (94dB SPL) incident pressure
- **dynamic range** → difference between maximal microphone SPL admitting 0,5%-3% THD (constructor measurements) and the noise floor in dBA
- **S/N** → difference between the reference pressure of 94dB SPL and the noise floor in dBA
- **transient response** capsule & electrical circuit response to transients
- **frequency response curve**



Technical specifications

Polar pattern:	cardioid
Frequency range:	20-20,000 Hz
Open circuit sensitivity:	10 mV/Pa (-40 dBV)
Maximum SPL (k = 1% / 3%):	132 / 142 dB
Equivalent noise level:	17 dB-A
Signal-to-noise ratio (A-weighted):	77 dB
Pre-attenuation pad:	-10 dB, switchable
Bass roll-off filter:	12 dB/octave at 75 Hz
Impedance:	200 Ω
Recommended load impedance:	> 1,000 Ω
Powering:	9-52 volts phantom power
Current consumption:	< 2 mA
Connector:	3-pin XLR plug
Finish:	dark grey
Dimensions:	19 Ø x 146 mm
Net / shipping weight:	115 / 420 g
Supplied accessories:	SA 40, W 90
Optional accessories:	B 18, H 30, H 38, H 300, SA 18/1B, SA 38/H, ST 1, ST 45, ST 305, MK 9/10, W 95

https://schoeps.de/en/knowledge/knowledge-base/technical-basics/pickup-pattern-and-the-polar-diagram.html

https://www.dpamicrophones.com/mic-university/technology/how-to-read-microphone-specifications/

Example. Shure SM 57

The impedance setting of a **mic preamp** should be about 10 times the actual impedance of the microphone ~10KΩ.

Rated Impedance:
- Low impedance <600 Ω
- Medium impedance 600 Ω -10 KΩ
- High impedance > 10KΩ

Handbook for Sound Engineers. Ch.21 .Glenn M. Ballou. Ed. Focal Press, 2002

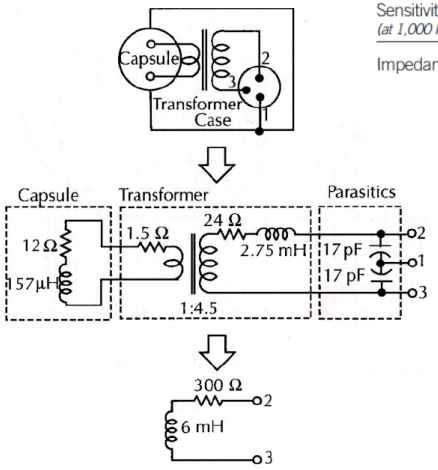


Figure 21-1. Shure SM57 schematic and equivalent circuits.

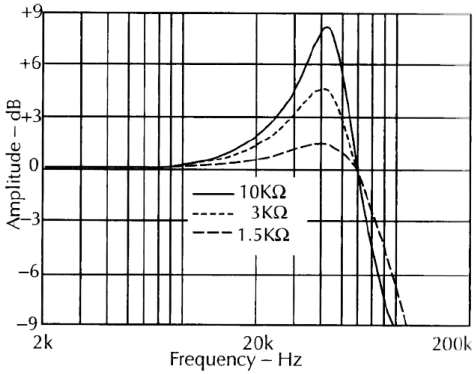


Figure 21-5. Effect of load resistance on damping.

Type	Dynamic
Frequency Response	40 to 15,000 Hz
Polar Pattern	Cardioid
Sensitivity *(at 1,000 Hz Open Circuit Voltage)*	Open Circuit Voltage: -56.0 dBV/Pa (1.6 mV) *(1 Pa = 94 dB SPL)*
Impedance	Rated impedance is 150Ω (310Ω actual) for connection to microphone inputs rated low impedance.

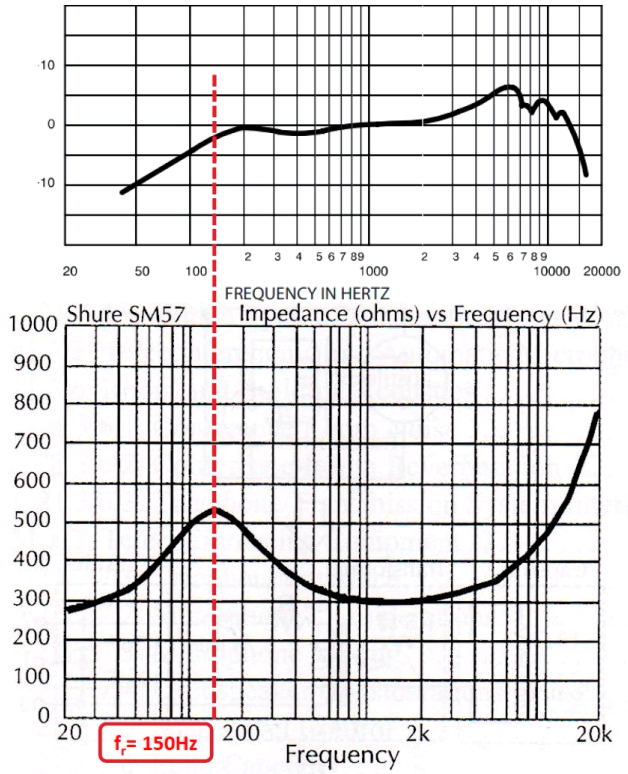
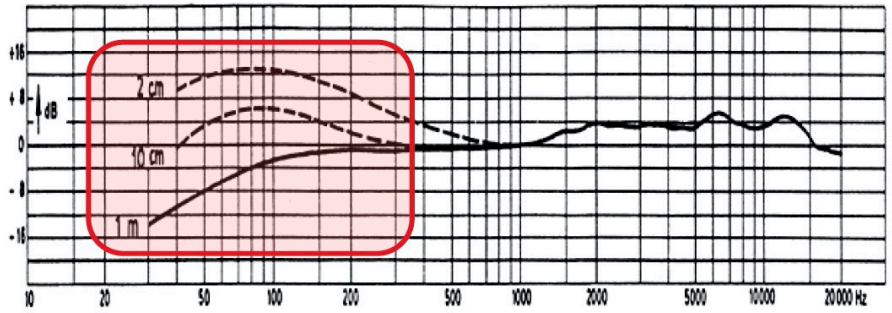


Figure 21-2. Measured output impedance of the Shure SM57.

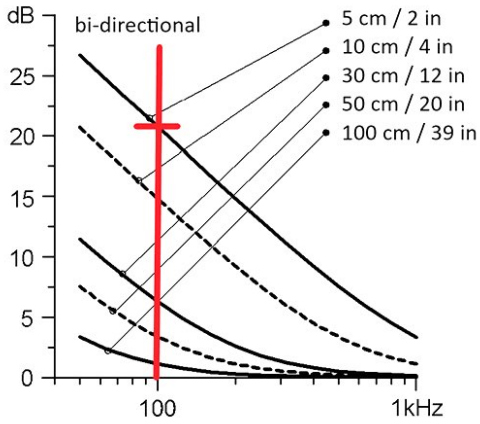
Proximity effect

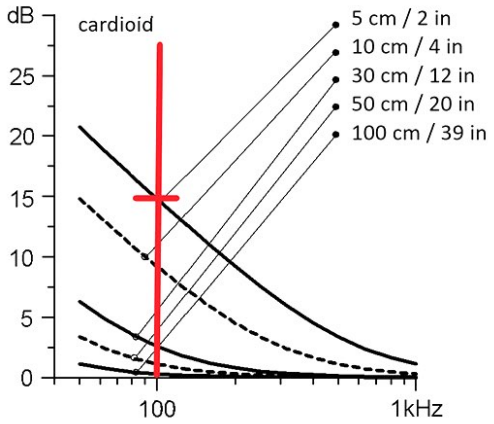
- implies a progressive increase in transducted low frequency energy (<200Hz) with the distance reduction between sound source and microphone capsule (cm)
- relevant for any directive microphone (order 1 or pressure gradient), *independently of transduction type* (electrostatic or electrodinamic)
- **not present** in omnidirectional mics (for any transduction type)



Higher in Bi-directional than in cardioid mics

https://www.dpamicrophones.com/mic-university/background-knowledge/proximity-effect-in-microphones-explained/





Proximity effect

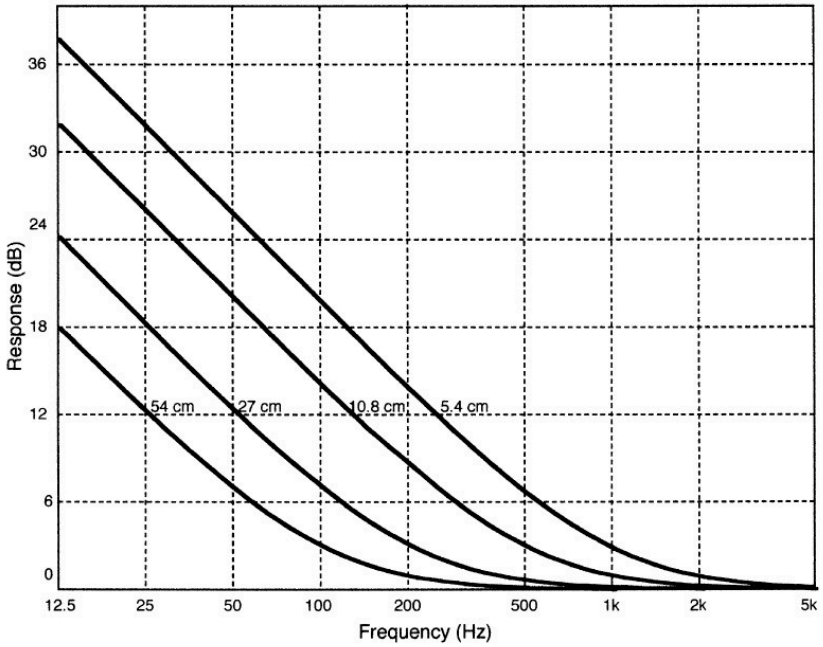


Figure 5.10
Proximity effect versus operating distance for a cardioid microphone on axis.

EARGLE, John.
The Microphone Book:
From mono to stereo to surround
-a guide to microphone design and application.
CRC Press, 2012, p 92.

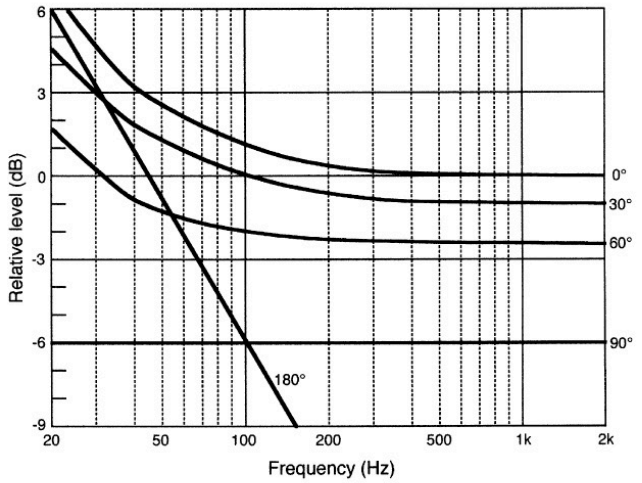
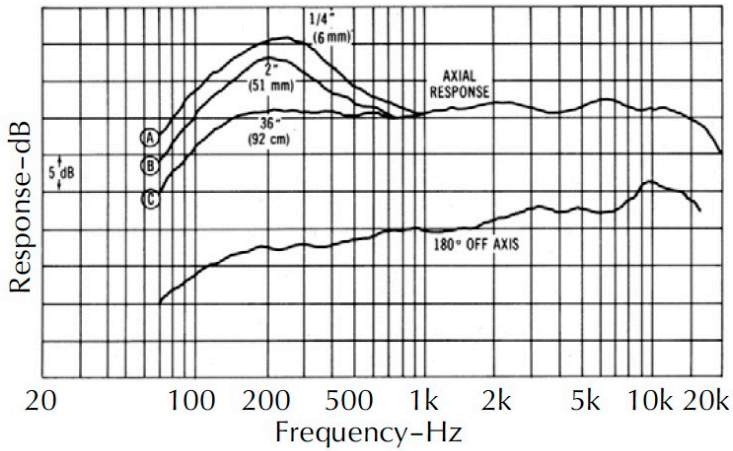


Figure 5.11
Proximity effect versus operating angle at a fixed distance of 0.6 m (24 in).

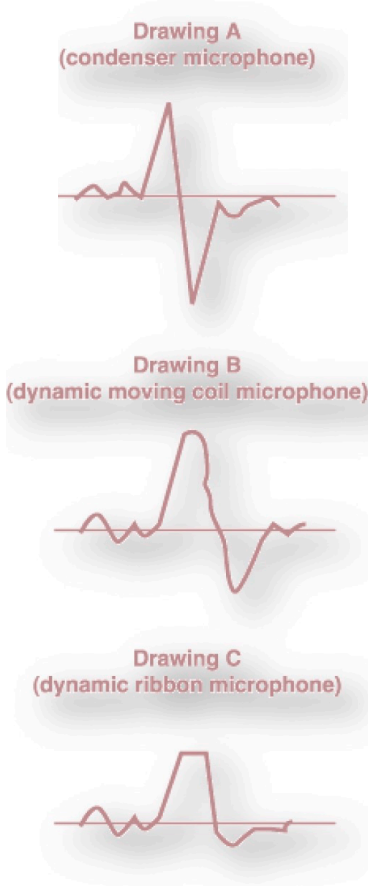


BALLOU, Glen.
Handbook for sound engineers.
Taylor & Francis, 2013, p 500.

Figure 16-21. Frequency response versus distance for an Electro-Voice DS35 single-entrant cardioid microphone. Courtesy Electro-Voice, Inc.

Transient response

- Ability to transduce sudden dynamic changes (pressure or velocity)



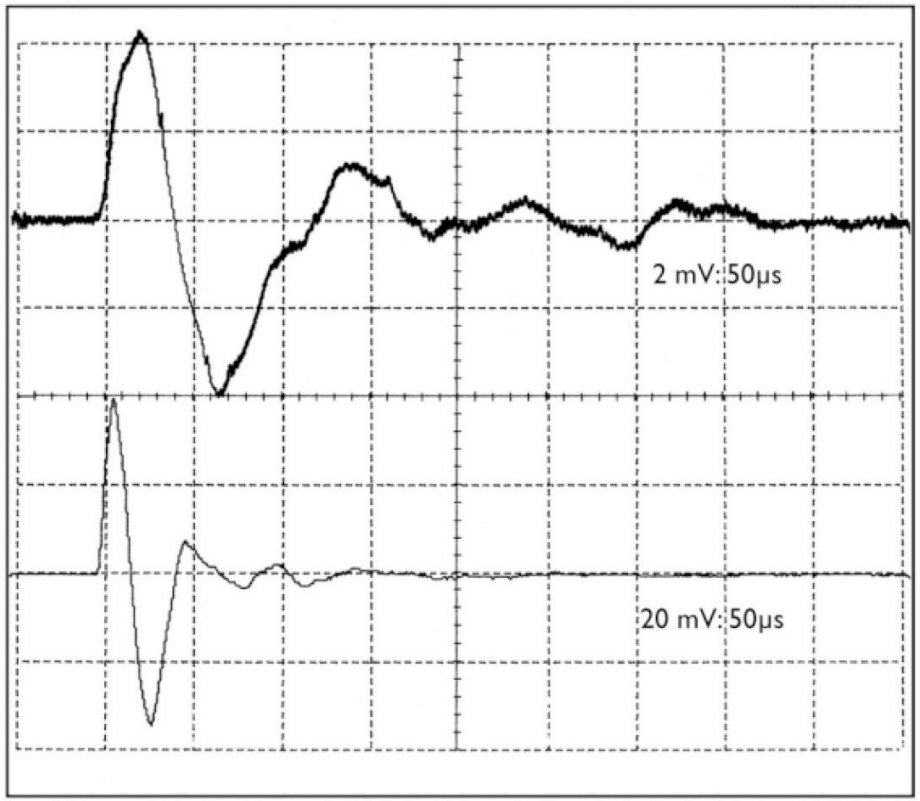


Fig. 41 Output voltages of two cardioid pattern studio microphones when stimulated by an electrical spark discharge (above: moving coil microphone, below: condenser microphone)

BORÉ, Gerhart; PEUS, Stephan. Mikrophone. Arbeitsweise und Ausführungsbeispiele, 1999, vol. 5., p66

Microphone source impedance

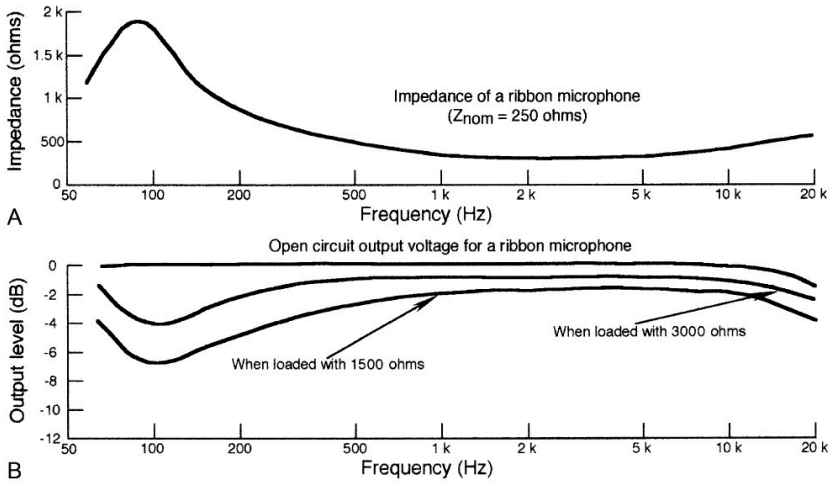


Figure 8.12
Effect of loading on ribbon microphone response.

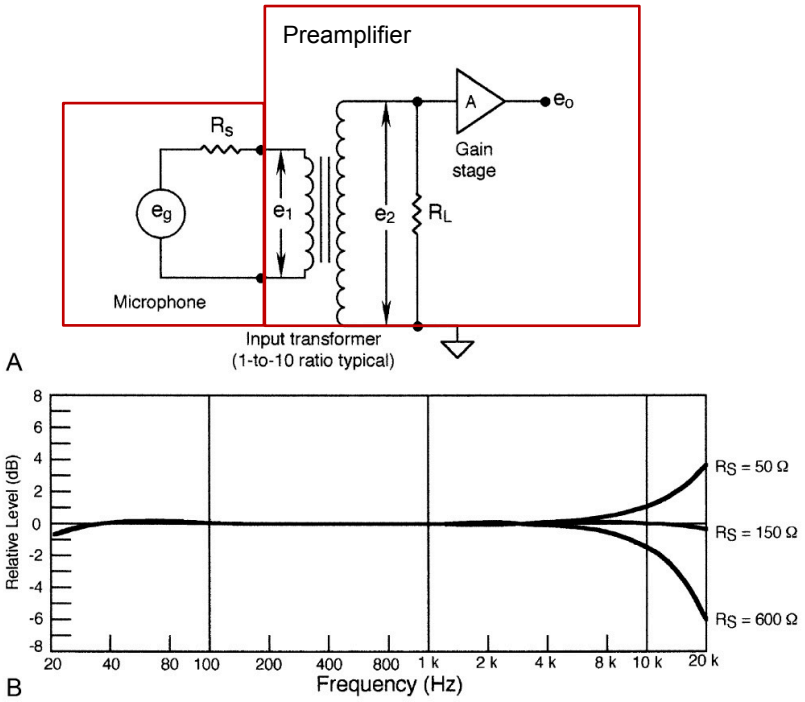


Figure 8.13
Response of transformer input circuit to varying values of microphone source impedance.

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, pp 158-159.

Inverse square law and Relative humidity

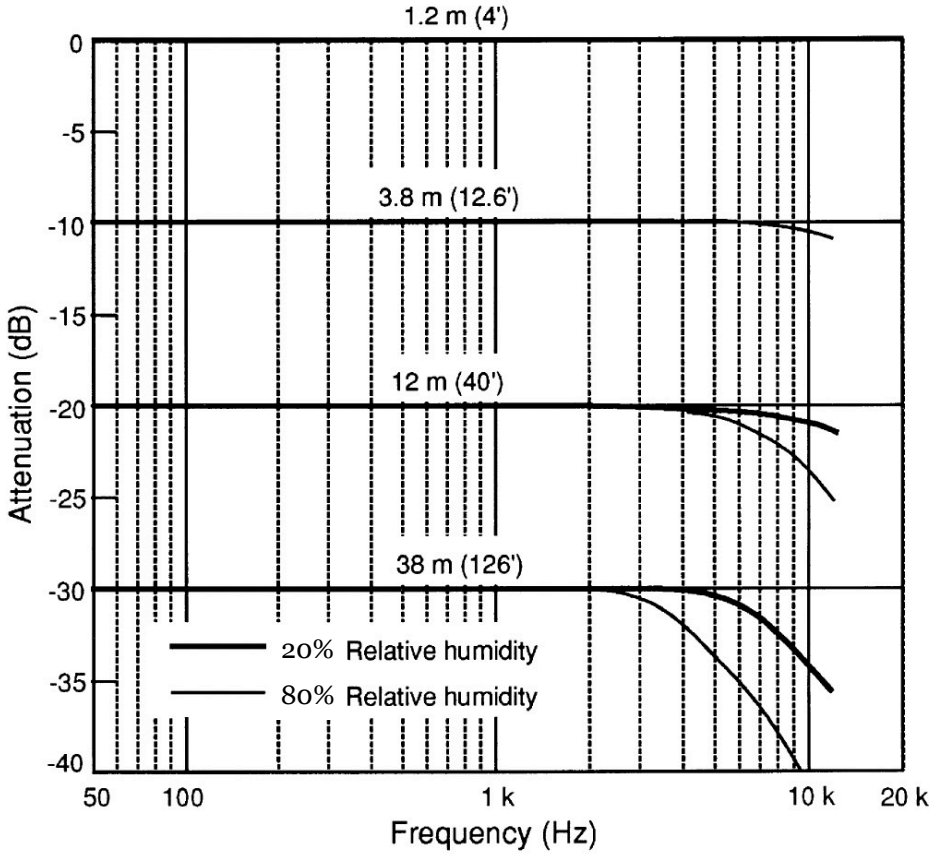


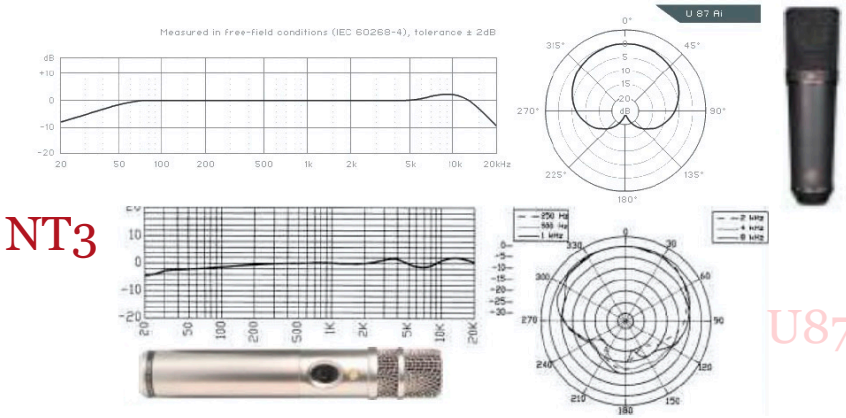
Figure 2.11
Effects of both inverse square relationships and HF air losses (20% and 80% RH).

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, p 22.

Comparative analysis

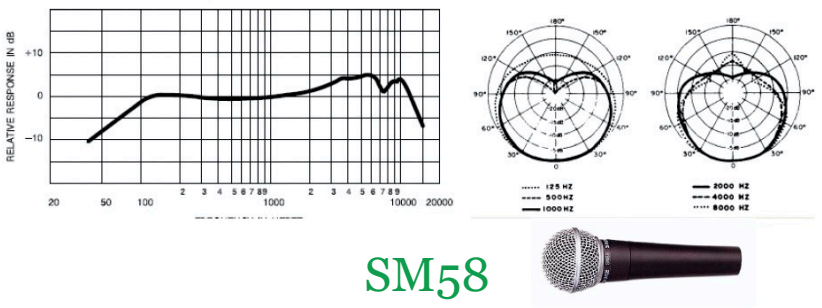
- U87 (1” diafragm) vs NT3 (electret) vs SM58 (moving-coil)

Transduction rule	electrostatic, pressure-gradient double membrane	back electret, pressure-gradient, single membrane	electrodynamic, pressure gradient, single membrane
capsule	1” (2,54cm)	¾” (1,9cm)	
directivity pattern	omnidirectional, cardioid, figure-of-eight	cardioid	cardioid
frequency response	20 Hz ... 20 kHz	20 Hz – 20 kHz	50Hz – 15KHz
sensitivity at 1KHz	20/28/22 mV/Pascal	12mv/Pascal	1,85mV/Pascal
output impedance	200Ω	200Ω	150Ω
equivalent floor noise level (dB-A)	15/12/14 dB-A	<16dB-A	
SNR, ref. 94 dB SPL (dB-A)	79/82/80 dB	>77dB-A	
maximum SPL	117 dB (cardioid),THD 0,5%	140 dB (cardioid), THD 1%	
dynamic range (dB-A)	105 dB	123dB	



NT3

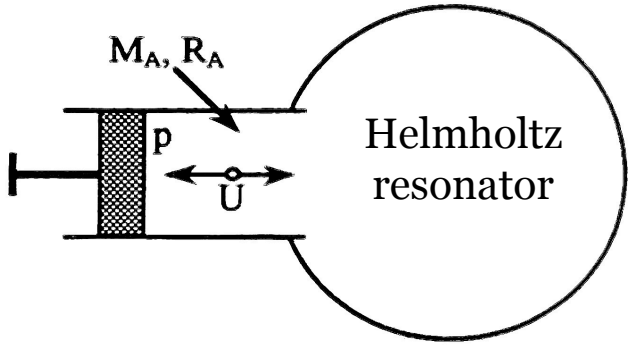
U87

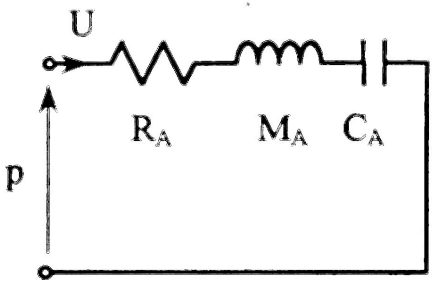


SM58

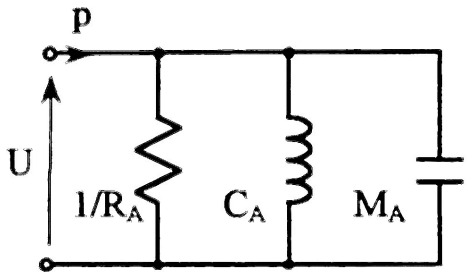
Electro-acoustic analogies (see additional slides)

The **compliance** of a gas is its property of changing in volume when subjected to an applied force. It is the reciprocal of stiffness.





Impedance analogy



Mobility analogy

Electric impedance: $V = I \,.\, Z \rightarrow Z = \frac{V}{I}$

Acoustic impedance:

$$Z_A = \frac{p}{U} = R_A + j\omega M_A + \frac{1}{j\omega C_A}$$

$[N.s/m^5]$

Electric	Z analogy	Mob. Analogy
Voltage	Pressure (p)	Flow (U)
Current	Flow (U)	Pressure (p)

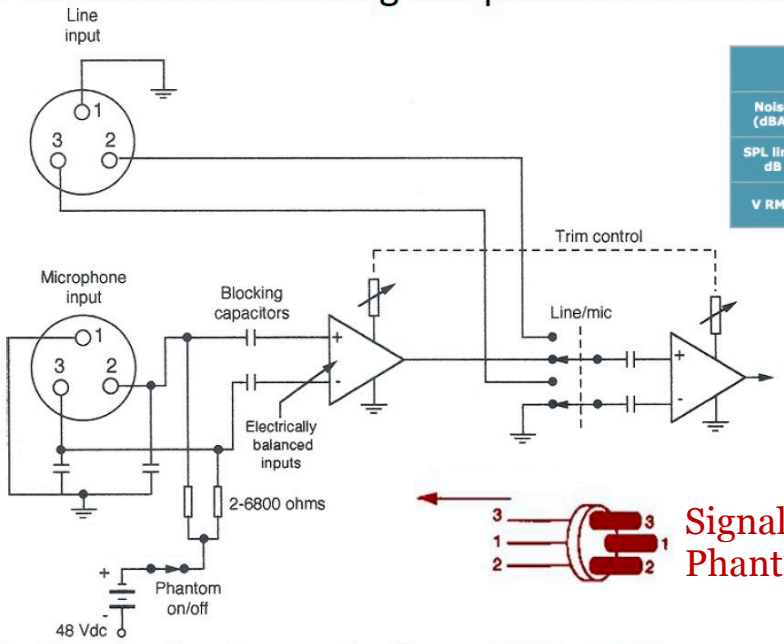
p (Pressure [Pa])

U = S.u $[m^3/s]$ (Volumetric Flow or volumetric velocity)

BOTE, José Luis Sánchez. *Transductores electroacústicos*. Fundación General UPM, ETSI Telecomunicación, 2014, p 20.

Phantom power

- direct current (DC) is needed to power up active transductors and DIs
  - phantom power is transmitted by the very same balanced audio cable
  - usually generated by a pre-amplifier in the mixing desk (48V) or a stand-alone battery (9V)
  - doesn't harm moving-coil passive transducers, but could make come off a ribbon membrane



	48 V	42 V	36 V	32 V	24 V	18 V	12 V	9 V
Noise (dBA)	26	26	26	26	26	26	26	N/A
SPL limit dB	142	142	142	142	136	134	126	N/A
V RMS	3	3	3	2.8	2.5	1.8	0.6	N/A

Signals go on pins 2(+) and 3(-)
Phantom power (48V DC) between pins 1-2 and 1-3

  - standard values rate from 9V to 54V
  - up to 200V "super phantom" (Josephson Engineering) powering allows for a current up to 43mA DC fed through a pair of 2K2Ω R and increases transduced voltage (sensitivity)

Phantom power

Mixing console with one phantom switch for all channels

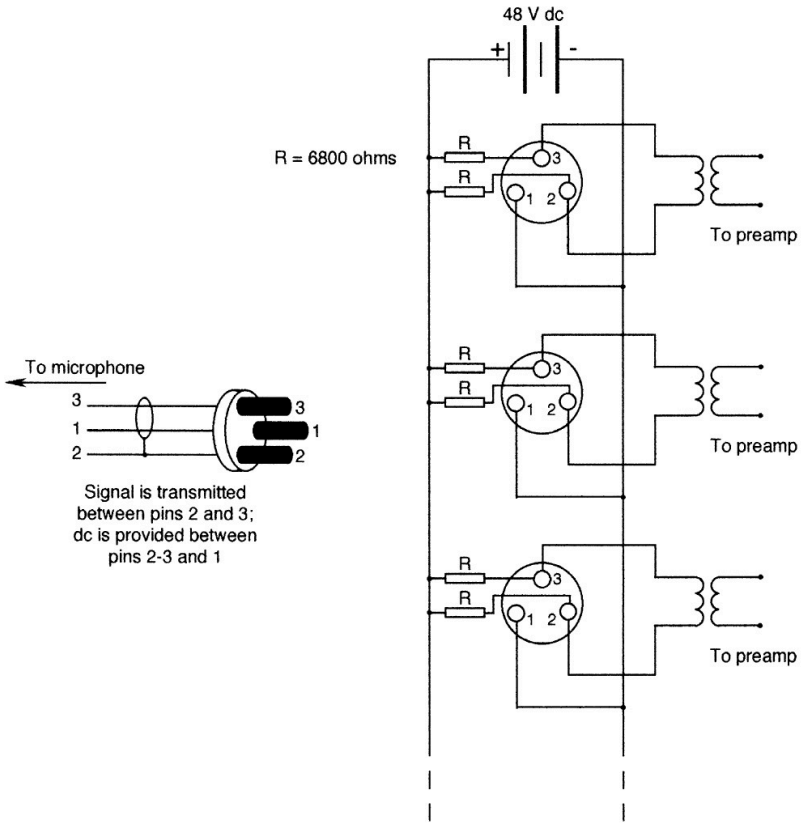


Figure 8.1
Basic circuit for phantom (simplex) powering.

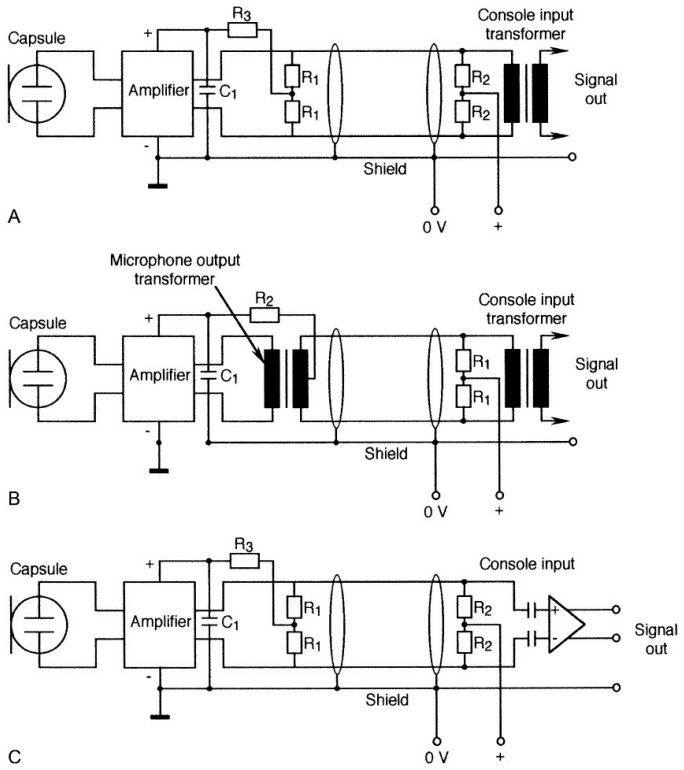


Figure 8.2
Microphone power input circuitry using a resistive voltage divider (A); using a center-tapped transformer secondary (B); using no transformers in the powering path (C).

Table 8.1 Voltage Tolerances and Current Limits

Designation	P12L	P12	P24	P48	SP48
Supply voltage	12 ±1 V	12 ±1 V	24 ± 4 V	48 ± 4 V	48 ± 4 V
Supply current	max. 8 mA	max. 15 mA	max. 10 mA	max. 10 mA	max. 22 mA
Rated supply current	4 mA	15 mA	10 mA	7 mA	22 mA
Feed resistors	3300 Ω	680 Ω	1200 Ω	6800 Ω	2200 Ω

EARGLE, John.
The Microphone Book: From mono to stereo to surround-a guide to microphone design and application.
CRC Press, 2012, pp 146-147.

12V A-B Powering (vintage equipment)

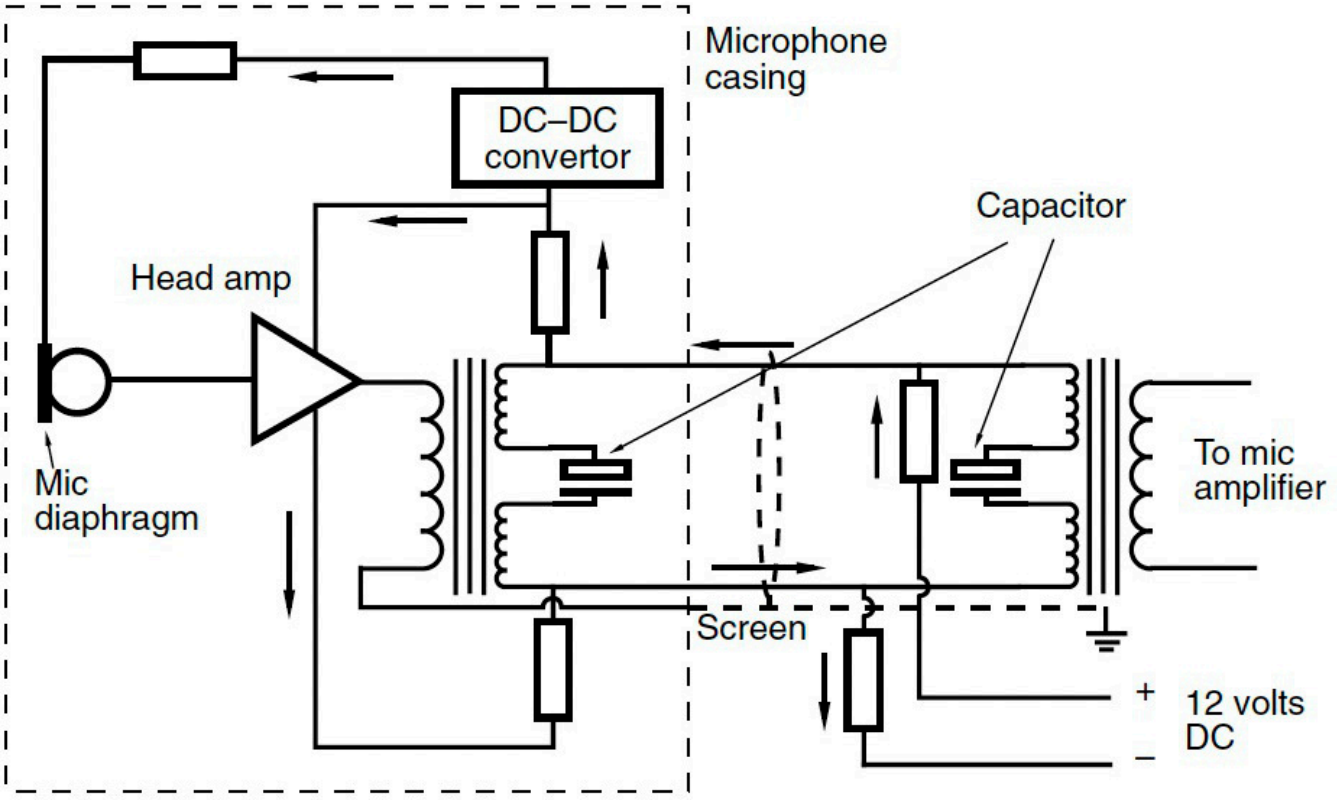
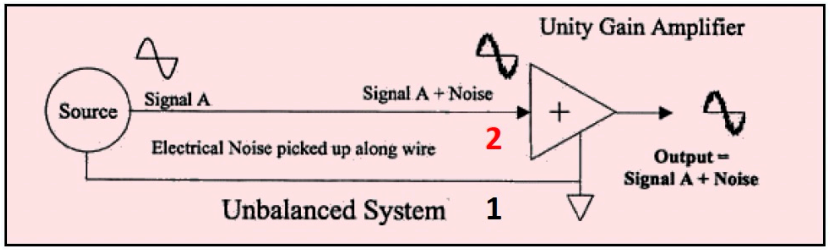


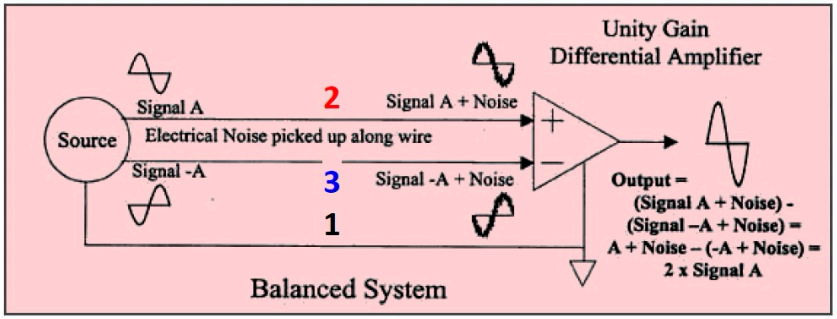
FIGURE 3.14 *A typical 12 volt A–B powering arrangement.*

RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012, p 69.

Balanced vs unbalanced. Audio connectors

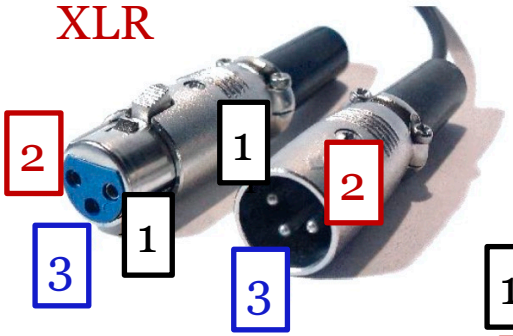
- balanced vs. non-balanced audio circuits

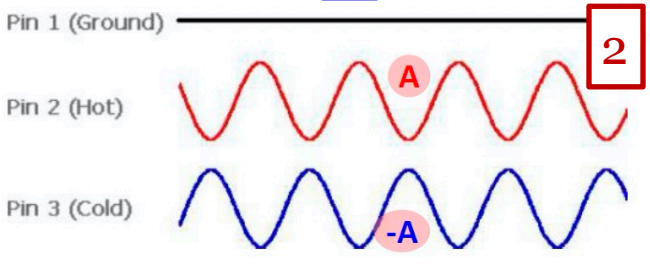




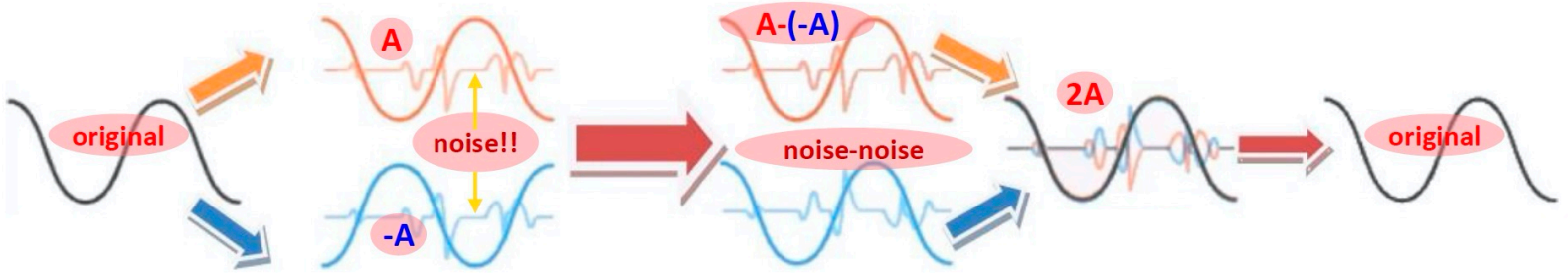
RCA TS

XLR





TRS

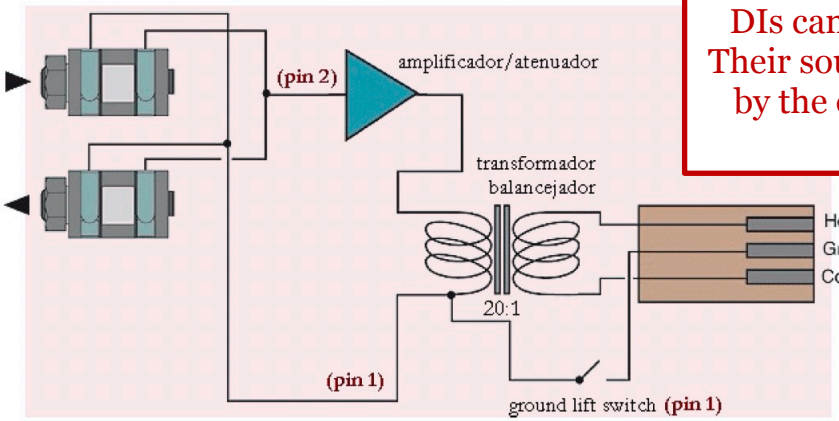


Phantom power needs a balanced connection

Direct Inputs (D.I.)

- transformate high output impedance (≈40KΩ) to low output impedance (≈100 Ω)
- transformate line-out level (≈1V) to microphone-out level (≈50mV)
- transformate a non-balanced to a balanced circuit
- allow *ground lift* from pin 1 (ground path is lifted in the DI-through connection)
- usually input signal atenuators and phase invertors Φ are also to be found

- longer cable lenghts are possible (less impedance means less EMIs)
- microphone-inputs in the console can be used to cope with line-inputs (keyboards, guitar)
- greter EMI refusal
- *hum* noise reduction induced by *ground loops*



DIs can be passive or active (Phantom). Their sound quality is respectively defined by the quality of the transformer or the electronic circuit.

Wireless Transmission

Analog FM

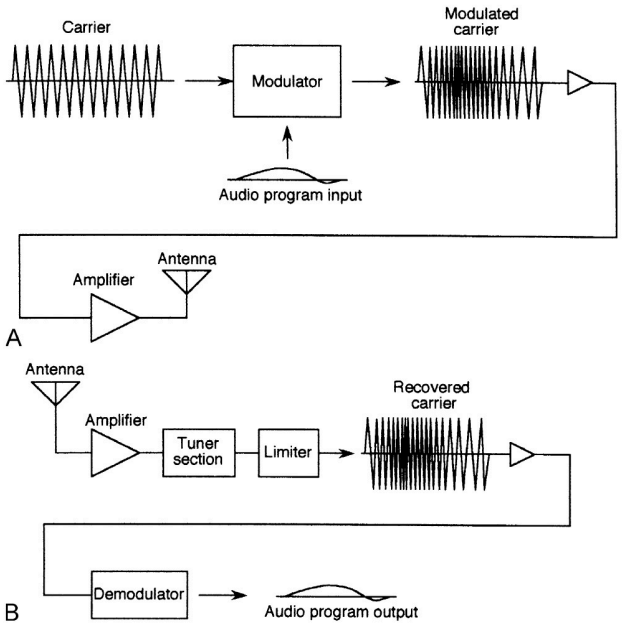


Figure 9.2
Principle of FM transmission and reception: simplified block diagram of a transmitter (A); simplified block diagram of a receiver (B).

Compander

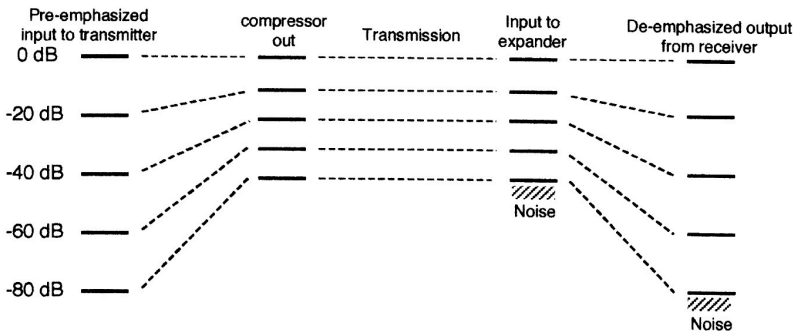


Figure 9.4
Principle of operation of signal companding (compression-expansion).

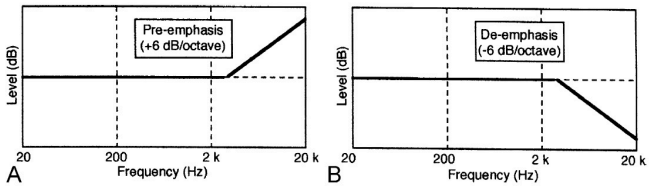


Figure 9.5
Typical HF pre-emphasis (A) and de-emphasis (B) curves used with wireless microphones.

Wireless basics: https://electrovoice.com/media/downloads/wireless_mic_basics.pdf

Digital wireless based on TDMA (Time Division Multiple Access):

https://en.wikipedia.org/wiki/Time-division_multiple_access

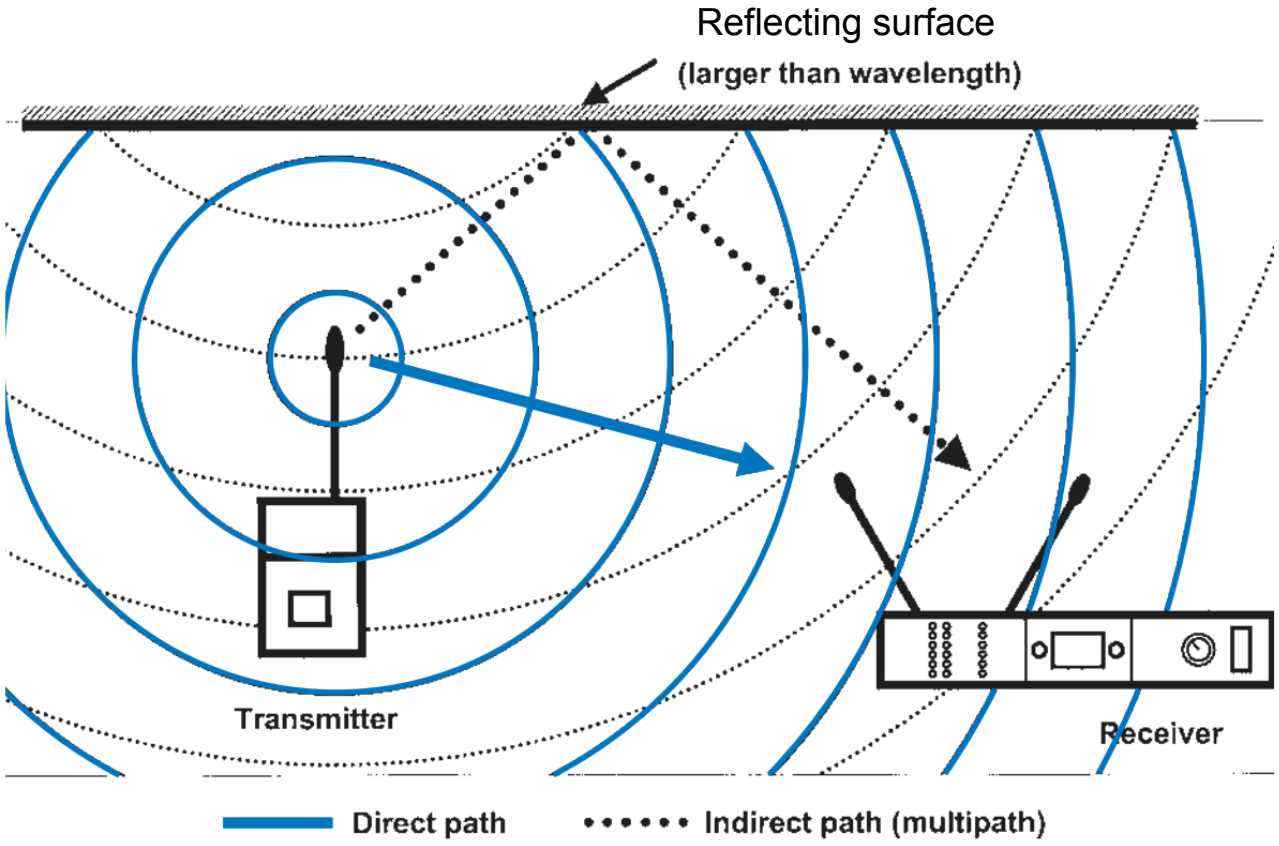
EARGLE, John.
The Microphone Book.From mono to stereo to surround -a guide to microphone design and application.
CRC Press, 2012, pp 184-186.

Diversity systems

https://www.shure.com

Diversity systems use multiple antennas to take advantage of the very low probability of simultaneous dropouts at two different antenna locations.

Figure 2-2: examples of transmitters (left to right: handheld, bodypack, plug-on)



Stereo orchestra recording

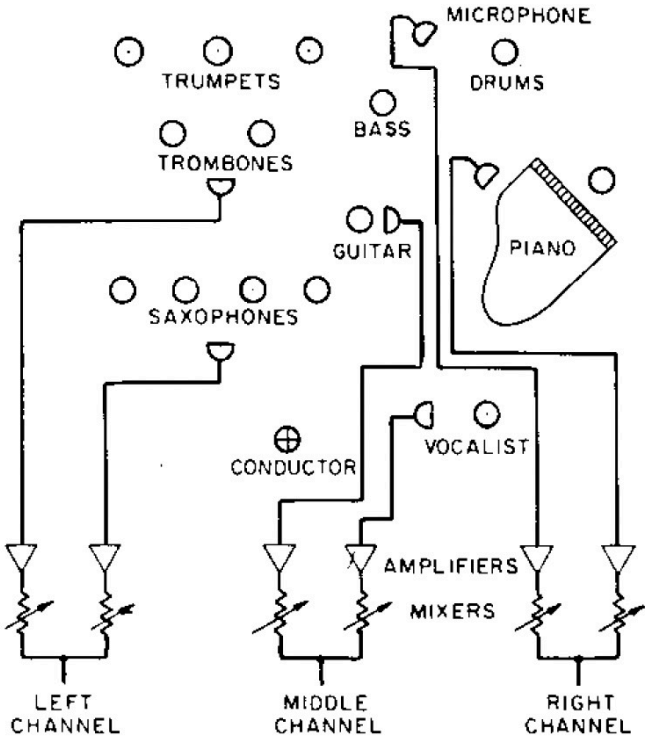


Fig. 19. Arrangement of the musical instruments, vocalist, and microphones for three-channel stereophonic sound recording of an orchestra.

OLSON, Harry F. Microphones for Recording. Journal of the Audio Engineering Society, 1977, vol. 25, no 10/11, p. 676-684.

Symphony Orchestra Recording for Interactive Opera Performances

Kanal/Mik nr	Multi	Sektion	Mikrofon	Mic Pre (Ch)	Interface (Ch)
1	3	DeccaL	4003	Millenna (1)	HD192 (1)
2	4	DeccaC	4003	Millenna (2)	HD192 (2)
3	5	DeccaR	4003	Millenna (3)	HD192 (3)
4	6	HS1L	U87	Presonus 1 (4)	HD192 (4)
5	7	HS1R	U87	Presonus 1 (5)	HD192 (5)
6	8	HS2L	U87	Presonus 1 (6)	HD192 (6)
7	9	HS2R	U87	Presonus 1 (7)	HD192 (7)
8	10	Vn1 (a)	4015	Yamaha (2)	HD192 (8)
9	11	Vn1 (b)	4015	Yamaha (4)	HD192 (9)
10	12	Vn1 (c)	USM69L	Presonus 2 (2)	HD192 (10)
11	13	Vn1 (d)	USM69R	Presonus 1 (1)	HD192 (11)
12	14	Vn2 (a)	4011	Presonus 2 (4)	HD192 (12)
13	15	Vn2 (b)	4011	Presonus 2 (5)	1296 (1)
14	16	Vn2 (c)	SM 69L	Presonus 2 (6)	1296 (2)
15	17	Vn2 (d)	SM 69R	Presonus 2 (7)	1296 (3)
16	18	Vla (a)	4011	Presonus 2 (8)	1296 (4)
17	19	Vla (b)	4011	Presonus 3 (1)	1296 (5)
18	20	Vlc (a)	U87	Presonus 1 (2)	1296 (6)
19	21	Vlc (b)	U87	Presonus 3 (3)	1296 (7)
20	22	Cb (a)	4006	Presonus 3 (4)	1296 (8)
21	23	Cb (b)	4006	Presonus 3 (5)	1296 (9)
22	24	Trb (a)	U87	Presonus 3 (6)	1296 (10)
23	25	Trb (b)	U87	Presonus 3 (7)	1296 (11)
24	26	Trp	U87	Presonus 3 (8)	1296 (12)
25	27	Hrn (a)	U87	M-Audio (1)	2408 Adat C (1)
26	28	Hrn (b)	U87	M-Audio (2)	2408 Adat C (2)
27	29	Tbl 1	Km84	M-Audio (3)	2408 Adat C (3)
28	30	Tbl 2	Km84	M-Audio (4)	2408 Adat C (4)
29	31	Tbl 3	Km84	M-Audio (5)	2408 Adat C (5)
30	32	Tbl 4	Km140	M-Audio (6)	2408 Adat C (6)
31	1	Hrn Solo	BLM3	M-Audio (7)	2408 Adat C (7)
32	2	Vn1 Solo	TLM103	M-Audio (8)	2408 Adat C (8)
33	3	Vlc Solo	TLM103	Yamaha (1)	2408 (1)
34	4			Octapre (2)	2408 (2)
35				Octapre (3)	2408 (3)
36				Octapre (4)	2408 (4)
37				Octapre (5)	2408 (5)
38				Octapre (6)	2408 (6)
39				Octapre (7)	2408 (7)
40				Octapre (8)	2408 (8)

Table 1: Microphones, preamplifiers and AD converters used. ("Kanal" = Channel, "Sektion" = Section, "Mikrofon" = Microphone)

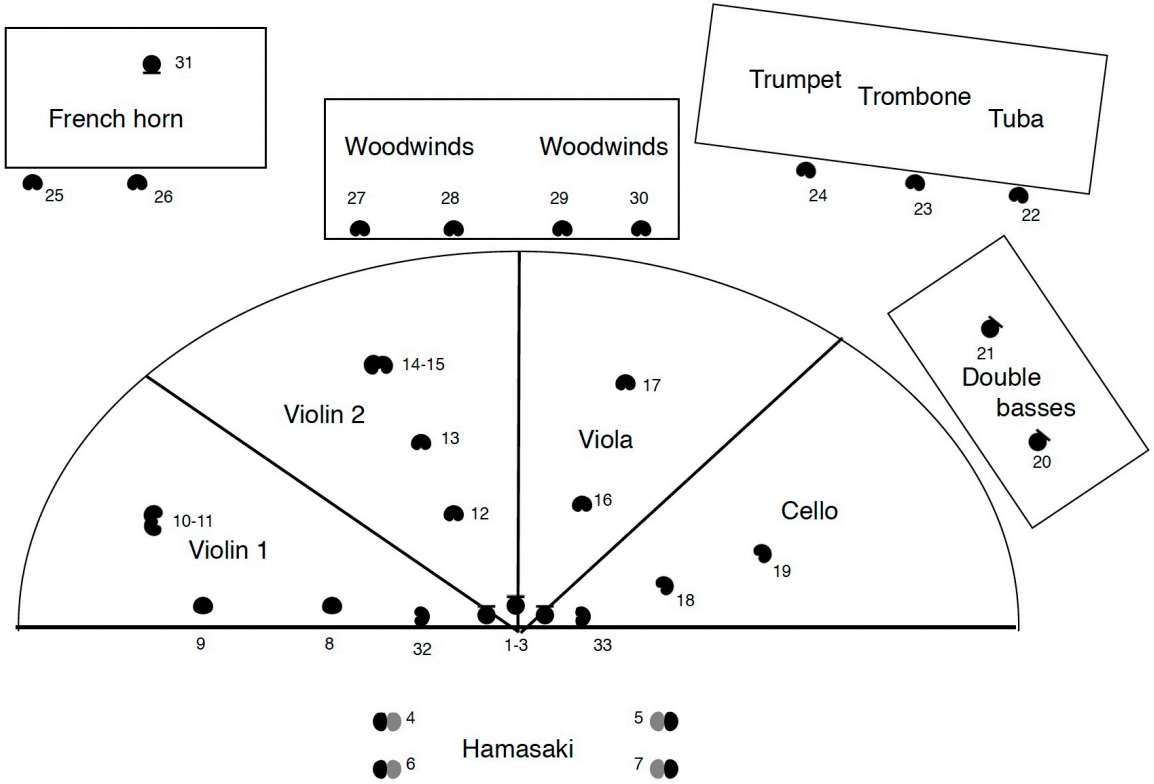


Figure 1: Microphone positions

HALLBERG, Lars; BERG, Jan. Symphony Orchestra Recording for Interactive Opera Performances. En *Audio Engineering Society Conference: 43rd International Conference: Audio for Wirelessly Networked Personal Devices*. Audio Engineering Society, 2011.

Level diagram

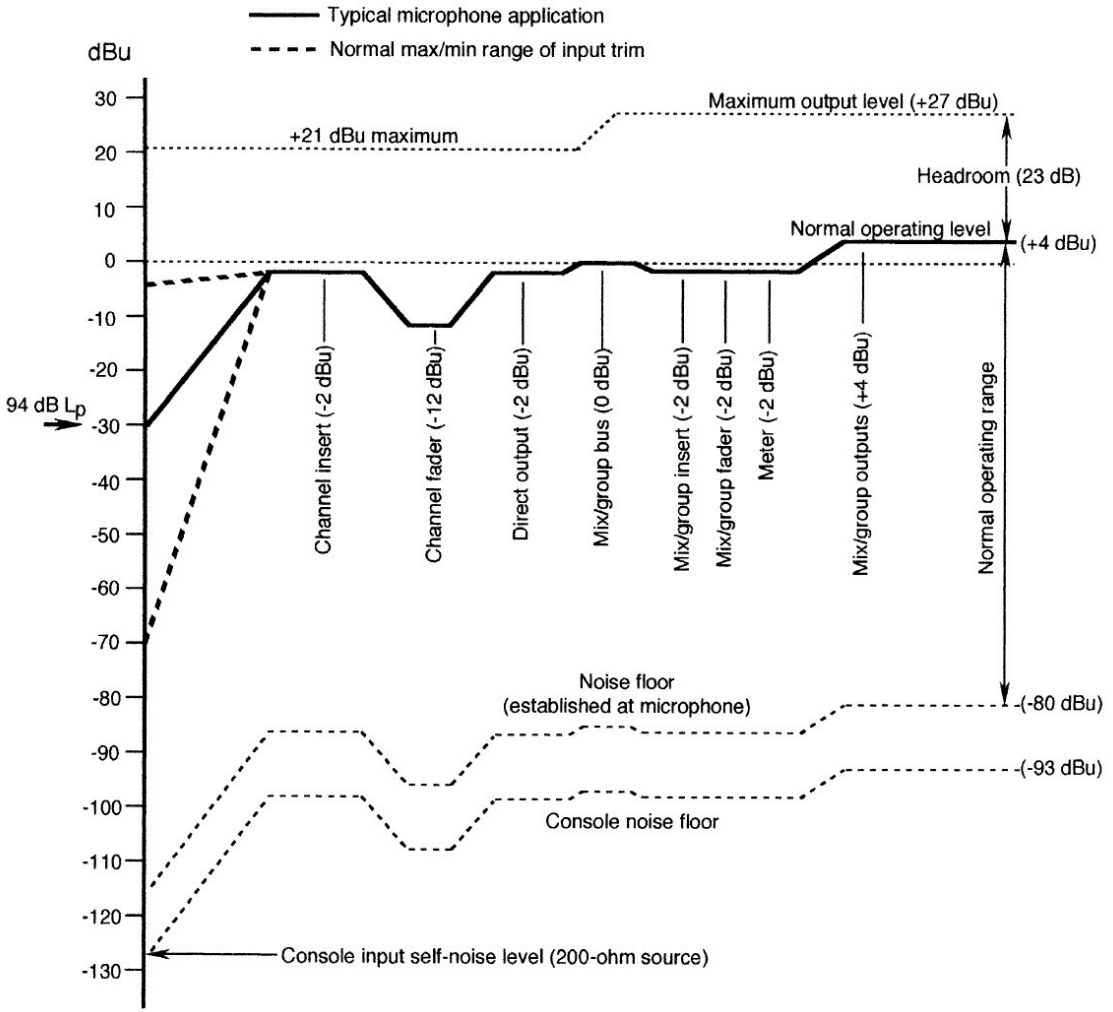


Figure 8.24
Typical console gain structure or level diagram.

EARGLE, John.
The Microphone Book.
From mono to stereo to surround
-a guide to microphone design and application.
CRC Press, 2012, p 173.

Notes about Timbre description

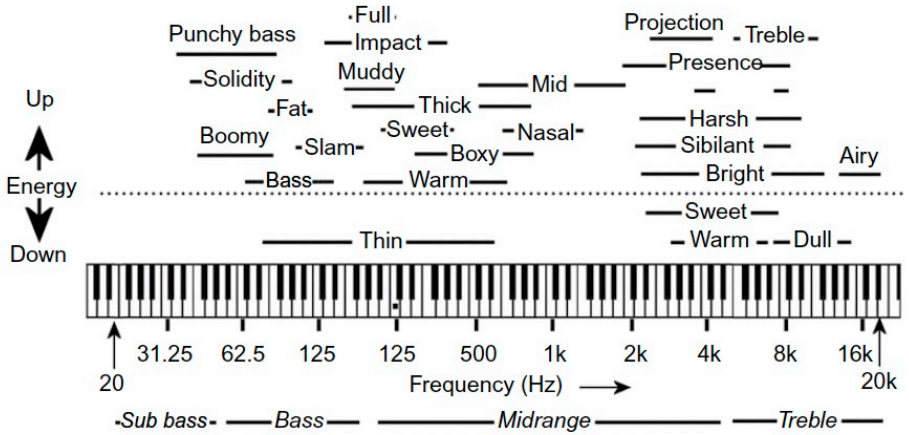


FIGURE 5.8 *Timbral descriptors used to describe the effect of boosting (above) and reducing (below) the spectral energy in various frequency regions set against a keyboard (middle C marked with a spot). Frequency values spanning the normally quoted human hearing range of 20 to 20 kHz, and terms commonly used to describe spectral regions are shown. (Adapted from Katz, 2007.)*

Table 5.1 Summary of the frequency domain properties as exemplified by the human hearing modeling spectrograms, example timbre descriptions and example acoustic instruments which fall under each of the four rules introduced by Helmholtz (1877). (Table adapted from Howard and Tyrrell, 1997)

Helmholtz rule	Human hearing modeling spectrogram	Example timbre descriptors	Example acoustic instruments
1	f_0 dominates	Pure Soft Simple Pleasant Dull at low pitch Free from roughness	Tuning fork Wide stopped organ flues Baroque flute
2	Harmonics dominate	Sweet and soft Rich Splendid Dark Dull Less shrill Bland	French horn, tuba Modern flute Recorder Open organ flues Soft sung sounds
3	Odd harmonics dominate	Hollow Nasal	Clarinet Narrow stopped organ flues
4	Striations dominate	Cutting Rough Bright Brilliant Shrill Brash	Oboe, bassoon Trumpet, trombone Loud sung sounds Bowed instruments Harmonium Organ reeds

HOWARD, David; ANGUS, Jamie. *Acoustics and psychoacoustics*. Routledge, 2013, pp 250-252.

Bibliography

QUIZ 3

BOOKS

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- RUMSEY, Francis; MCCORMICK, Tim. *Sound and recording: an introduction*. CRC Press, 2012.
- EARGLE, John. *The Microphone Book:From mono to stereo to surround-a guide to microphone design and application*.CRC Press, 2012.
- *HOWARD, David; ANGUS, Jamie. Acoustics and psychoacoustics. Routledge, 2013.*
- *BOTE, José Luis Sánchez. Transductores electroacústicos. Fundación General UPM, ETSI Telecomunicación, 2014.*

LINKS

- AEA: https://aearibbonmics.com/
- AKG: https://www.akg.com/Microphones
- Beyerdynamic : https://www.beyerdynamic.de/mikrofone.html
- Brüel & Kjaer: https://www.bksv.com/es/transducers
- Coles Electroacoustics: https://coleselectroacoustics.com/microphones/
- DPA: https://www.dpamicrophones.com/
- Electro-Voice: https://products.electrovoice.com/emea/en/wired-music-performance-microphones-pt/
- Josephson: https://www.josephson.com/
- Neumann: https://www.neumann.com/es-es/microphones/
- Royer Labs: https://royerlabs.com/microphones/
- Schoeps: https://schoeps.de/en/
- Sennheiser: https://es-mx.sennheiser.com/proaudio
- Shure: *https://techportal.shure.com/es/productos/microfonos*
- Sonotronics: https://www.sontronics.com/products
- Telefunken elektroakustik: https://www.telefunken-elektroakustik.com/

- Audio calculators: http://www.sengpielaudio.com/Calculations03.htm
- Feedback calculator: https://www.shure.com/en-US/support/tools/pag-nag-calculator

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